

Relaxation-Driven Cyclic Cosmology (RDCC)

Flagship Paper

Michael Lehmann¹

¹Independent Researcher, mi.lehmann@gmx.de

March 2026

Executive Summary

Relaxation-Driven Cyclic Cosmology (RDCC) is a minimal and predictive framework for a CPT-symmetric Universe. It combines three structural ingredients:

- (a) a bipartite Hilbert-space factorization $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$, relating the visible sector to its CPT-conjugate mirror sector,
- (b) a Planck-suppressed inter-sector operator that induces a small, universal relaxation parameter $\alpha \sim 10^{-2}$,
- (c) a cuscuton-induced, ghost-free nonsingular bounce that connects the contracting and expanding phases.

These ingredients yield a cosmology that is globally CPT-symmetric while reproducing the observed sectoral arrow of time, the dark-matter abundance, the scalar spectral tilt, and percent-level deviations from Λ CDM.

Global CPT Structure and the Arrow of Time

RDCC postulates that the full quantum state of the Universe is a CPT eigenstate. This implies a bipartite Hilbert-space structure in which each sector exhibits its own thermodynamic arrow of time, while the combined system satisfies

$$S_{\text{tot}} = S_+ + S_- = \text{const.}$$

The bounce corresponds to the hypersurface of minimal coarse-grained entropy. The relaxation parameter α quantifies the rate of entanglement leakage between the sectors and flows to zero at the bounce, ensuring maximal correlation and restoring manifest CPT symmetry.

Effective Field Theory and the Origin of α

The inter-sector coupling is generated by the unique dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu},$$

which arises naturally from integrating out heavy gravitationally coupled fields. Its renormalization-group flow exhibits an infrared fixed point,

$$\alpha(\mu) \longrightarrow \alpha_* \sim 10^{-2},$$

providing a universal explanation for the magnitude of the relaxation parameter. Microscopically, α admits three complementary interpretations: (i) a loop-suppressed coefficient, (ii) an entanglement-leakage rate, and (iii) an RG-attractor controlling sectoral equilibration.

Gravity as the Global Bridge

Gravity is the only interaction that couples to the full CPT-symmetric state. The global Einstein equations involve the sum of the two stress-energy tensors,

$$G_{\mu\nu} = 8\pi G_N (T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)}),$$

while a visible-sector observer perceives an effective Newton constant

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+}.$$

The mirror sector therefore acts as a *gravitational shadow*, providing a structural explanation for dark matter without introducing new particle species.

Bounce Dynamics and Phase-Space Stability

The bounce is generated by a cusciton-like operator that produces controlled NEC violation without propagating additional degrees of freedom. The modified Friedmann equation takes the effective form

$$H^2 = \frac{\rho_{\text{tot}}}{3M_{\text{Pl}}^2} \left(1 - \frac{\rho_{\text{tot}}}{\rho_{\text{crit}}} \right),$$

ensuring a smooth turnaround at $\rho_{\text{tot}} = \rho_{\text{crit}}$. A phase-space analysis shows that the CPT-symmetric trajectory is a dynamical attractor: antisymmetric perturbations between the sectors are damped by the relaxation dynamics, and the bounce is stable for a wide range of initial data.

Perturbations and Gravitational Waves

RDCC predicts a nearly scale-invariant scalar spectrum with

$$n_s - 1 = -2\alpha,$$

and a blue-tilted stochastic gravitational-wave background peaked in the millihertz range. A distinctive signature is the presence of log-periodic oscillations in the tensor spectrum,

$$\Omega_{\text{GW}}(k) = \Omega_0(k) \left[1 + \varepsilon \cos \left(\omega \ln \frac{k}{k_b} + \phi \right) \right],$$

arising from interference between the two CPT-related tensor sectors across the bounce. These oscillations provide a clear observational target for LISA.

Phenomenology and Falsifiability

All deviations from Λ CDM are governed by the single parameter α . RDCC predicts correlated signatures in:

- the scalar spectral tilt,
- the dark-radiation excess ΔN_{eff} ,
- the growth-rate observable $f\sigma_8(z)$,
- the dark-matter to baryon ratio,

- the stochastic gravitational-wave background.

The theory is sharply falsifiable: any inconsistency among these correlated predictions would rule out RDCC, while detection of log-periodic GW modulations would strongly support it.

Abstract

Fixing the relaxation parameter from the observed scalar tilt yields $\alpha \simeq 0.0175$, in excellent agreement with the expected infrared fixed-point value. This single parameter governs all correlated deviations from Λ CDM in Relaxation-Driven Cyclic Cosmology (RDCC), providing a direct observational anchor for the framework. In the accompanying observational analysis (Companion V), we present a first prototype global fit combining representative constraints from the scalar tilt, dark radiation, large-scale structure, and upper bounds on the stochastic gravitational-wave background. The resulting posterior peaks at $\alpha \sim 0.016$, fully consistent with the analytic estimate and demonstrating that the correlated RDCC prediction structure is compatible with current data at the present level of precision.

RDCC offers a minimal and predictive realization of a CPT-symmetric Universe. The construction is based on three structural ingredients: (i) a bipartite Hilbert-space factorization relating the visible sector to its CPT-conjugate mirror sector, (ii) a Planck-suppressed inter-sector operator that induces the universal relaxation parameter α , and (iii) a cusp-ton-induced, ghost-free nonsingular bounce that connects the contracting and expanding phases within the regime of validity of effective field theory.

The global CPT structure implies that the total entropy of the two-sector system is conserved, while each sector exhibits its own thermodynamic arrow of time. The bounce corresponds to the hypersurface of minimal coarse-grained entropy, where $\alpha \rightarrow 0$ and the two sectors become maximally correlated. Gravity acts as the unique global interaction, coupling to the sum of the two stress-energy tensors and giving rise to an effective Newton constant $G_{\text{eff}} = G_N/(1 + \rho_-/\rho_+)$. The mirror sector therefore appears as a gravitational shadow, providing a structural explanation for dark matter without introducing new particle species.

The relaxation parameter originates from the renormalization-group flow of the inter-sector operator and admits complementary microscopic interpretations as a loop-suppressed coefficient, an entanglement-leakage rate, and an infrared fixed point. The bounce is dynamically stable: a phase-space analysis shows that antisymmetric perturbations between the sectors are damped, and the CPT-symmetric trajectory acts as an attractor.

RDCC predicts a nearly scale-invariant scalar spectrum with $n_s - 1 = -2\alpha$, a blue-tilted stochastic gravitational-wave background peaked in the millihertz range, and distinctive log-periodic oscillations in the tensor spectrum arising from interference between the two CPT-related sectors. All deviations from Λ CDM follow from the single parameter α , yielding a sharply falsifiable, one-parameter correlation structure across early- and late-time cosmology.

1 Introduction

The complete Relaxation-Driven Cyclic Cosmology (RDCC) framework, consisting of this Flagship Paper and all 12 Companion papers, is openly available as a single deposit on Zenodo: <https://zenodo.org/records/19580659>.

The standard Λ CDM model provides an exceptionally successful phenomenological description of the expansion history and the statistics of cosmological perturbations. Yet several foundational questions remain unresolved within this framework. These include the origin of the thermodynamic arrow of time, the nature of the initial singularity, the Tolman entropy problem, the structural origin of dark matter, and the dynamical mechanism underlying the baryon asymmetry. Existing proposals typically address these puzzles through independent extensions of the standard cosmological model—such as inflation, modified gravity, or new particle species—without explaining why these issues arise together or why their proposed solutions remain uncorrelated.

Relaxation-Driven Cyclic Cosmology (RDCC) offers a minimal and internally consistent completion of Λ CDM that addresses these questions within standard General Relativity and effective field theory (EFT). The framework is built on a single structural postulate: the global quantum state of the Universe is a CPT eigenstate. The minimal EFT-level realization of such a state requires a bipartite Hilbert-space factorization,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-,$$

where $\mathcal{H}_+$ describes the visible sector and $\mathcal{H}_-$ its CPT-conjugate mirror sector. The two sectors propagate on the same spacetime metric and interact only through gravity and a Planck-suppressed irrelevant operator. This construction introduces no new fundamental fields; instead, it specifies the global structure of the cosmological state.

A defining feature of RDCC is its one-parameter structure. A single infrared relaxation parameter $\alpha \sim 10^{-2}$, generated by the leading dimension-six inter-sector operator, governs correlated deviations across multiple cosmological epochs. These include:

- the scalar spectral tilt $n_s - 1 = -2\alpha$,
- the dark-radiation excess $\Delta N_{\text{eff}} \sim \alpha$,
- percent-level corrections to the growth-rate observable $f\sigma_8(z)$,
- the dark-matter to baryon ratio,
- the amplitude and structure of the stochastic gravitational-wave background.

The relaxation parameter admits complementary microscopic interpretations: as a loop-suppressed coefficient of the inter-sector operator, as a measure of entanglement leakage between the sectors, and as an infrared fixed point of the renormalization-group flow. These interpretations reinforce the universality of α and explain why a single dimensionless parameter governs correlated deviations from Λ CDM.

A second structural ingredient of RDCC is the cuscuton-induced nonsingular bounce. The cuscuton arises naturally from integrating out heavy gravitationally coupled fields and provides a controlled, ghost-free mechanism for violating the null energy condition. The resulting modified Friedmann equation,

$$H^2 = \frac{\rho_{\text{tot}}}{3M_{\text{Pl}}^2} \left(1 - \frac{\rho_{\text{tot}}}{\rho_{\text{crit}}} \right),$$

ensures a smooth transition between contraction and expansion at the critical density ρ_{crit} . A phase-space analysis shows that the CPT-symmetric trajectory is a dynamical attractor: antisymmetric perturbations between the sectors are damped by the relaxation dynamics, and the bounce is stable for a wide range of initial conditions.

The third structural ingredient is the role of gravity as the unique global interaction. While gauge interactions act strictly within a single sector, the metric couples to the sum of the two stress-energy tensors,

$$G_{\mu\nu} = 8\pi G_N (T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)}).$$

A visible-sector observer therefore perceives an effective Newton constant,

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+},$$

and infers the presence of an additional gravitational component. This *gravitational shadow* of the mirror sector behaves exactly like cold dark matter, providing a structural explanation for the observed dark-matter abundance without introducing new particle species.

RDCC predicts a nearly scale-invariant scalar spectrum, a blue-tilted stochastic gravitational-wave background peaked in the millihertz range, and distinctive log-periodic oscillations in the tensor spectrum arising from interference between the two CPT-related sectors across the bounce. These oscillations provide a clear observational target for LISA and constitute a smoking-gun signature of the RDCC framework.

Structure of This Paper

This paper presents the flagship formulation of RDCC. Section 2 develops the global CPT structure, the bipartite Hilbert-space factorization, and the thermodynamic interpretation of the arrow of time. Section 3 constructs the effective field theory, including the origin and RG flow of the relaxation parameter and the microscopic derivation of the cusciton operator. Section 4 develops the global and sectoral Einstein equations, derives the effective Newton constant, and defines the gravitational shadow of the mirror sector. Section 5 analyzes the bounce dynamics and phase-space stability. Section 6 develops the scalar and tensor perturbations, including the analytical origin of log-periodic gravitational-wave oscillations. Section 7 summarizes the phenomenological predictions and falsifiability criteria. We conclude with a discussion of open questions and future directions.

Notion of Minimality. The term “minimal” refers to the absence of new propagating degrees of freedom; the global two-sector structure is a structural rather than dynamical extension.

Nomenclature

List of Symbols

Table 1: List of Symbols used in this work.

Symbol	Description	First Appearance
ϕ	Effective Higgs-like scalar field driving the background dynamics	Sec. 3
$V_{\text{eff}}(\phi)$	Radiatively generated scalar potential	Sec. 3
M_{Pl}	Reduced Planck mass	Sec. 2
M_*	Heavy UV scale generating the inter-sector operator and cusciton term	Sec. 3
$T_{\mu\nu}^{(\pm)}$	Stress-energy tensors of visible (+) and mirror (−) sectors	Sec. 2
$\mathcal{H}_{\pm}$	Visible and mirror Hilbert-space sectors	Sec. 2
α	Infrared relaxation parameter controlling inter-sector coupling	Sec. 3
$\beta(\alpha)$	Beta function governing RG flow of α	Sec. 3
ρ_{tot}	Total energy density of both sectors	Sec. 4
p_{tot}	Total pressure of both sectors	Sec. 4
ρ_{crit}	Critical density at which the bounce occurs	Sec. 4
H	Hubble parameter ($H = \dot{a}/a$)	Sec. 4
$a(t)$	Scale factor	Sec. 4
c_s^2	Effective sound speed of scalar perturbations	Sec. 4
$\mathcal{M}$	Stability matrix of the background system (phase-space analysis)	Sec. 5
$x = T'/T$	Mirror-to-visible temperature ratio	Sec. 7
Y'_{He}	Mirror helium mass fraction	Sec. 7
Ω_{DM}	Dark matter density parameter (mirror baryons)	Sec. 7
Ω_b	Visible baryon density parameter	Sec. 7
$f\sigma_8(z)$	Growth-rate observable	Sec. 7
n_s	Scalar spectral index ($n_s = 1 - 2\alpha$)	Sec. 6
ΔN_{eff}	Dark-radiation excess from mirror sector	Sec. 6
$\Omega_{\text{GW}}(f)$	Stochastic gravitational-wave spectrum	Sec. 6
f_{peak}	Peak GW frequency in the LISA band	Sec. 6
ζ	Curvature perturbation	Sec. 6
$\delta\phi$	Scalar perturbation of ϕ	Sec. 6
$K(t), G(t)$	Kinetic coefficients in quadratic action	Sec. 6

List of Acronyms and Abbreviations

Table 2: List of Acronyms used in this work.

Acronym	Description	First Appearance
RDCC	Relaxation-Driven Cyclic Cosmology	Title
CPT	Charge-Parity-Time symmetry	Sec. 2
EFT	Effective Field Theory	Sec. 3
RG	Renormalization Group	Sec. 3
BBN	Big Bang Nucleosynthesis	Sec. 7
CMB	Cosmic Microwave Background	Sec. 6

Acronym	Description	First Appearance
LSS	Large-Scale Structure	Sec. 6
SGWB	Stochastic Gravitational-Wave Background	Sec. 6
LISA	Laser Interferometer Space Antenna	Sec. 6
DESI	Dark Energy Spectroscopic Instrument	Sec. 6
Euclid	Euclid Space Telescope	Sec. 6
LSST	Legacy Survey of Space and Time	Sec. 6
SKA	Square Kilometre Array	Sec. 6
FLRW	Friedmann–Lemaître–Robertson–Walker metric	Sec. 4
SM	Standard Model	Sec. 8

2 Foundations of the RDCC Framework

Relaxation-Driven Cyclic Cosmology (RDCC) is built on a structural postulate: the full quantum state of the Universe is a CPT eigenstate. This assumption does not constrain local microphysics directly; instead, it fixes the global structure of the cosmological state and determines how the thermodynamic arrow of time, the bounce, and the inter-sector dynamics emerge from a single underlying principle.

In this section we develop the foundational ingredients of RDCC: the global CPT postulate, the bipartite Hilbert-space factorization, the thermodynamic interpretation of the arrow of time, and the structural role of the relaxation parameter α .

2.1 Global CPT Postulate

The central structural assumption of RDCC is that the full quantum state $|\Psi\rangle$ of the Universe satisfies

$$\widehat{\text{CPT}} |\Psi\rangle = |\Psi\rangle. \quad (1)$$

This condition is consistent with the general expectation that CPT symmetry is exact in any local Lorentz-invariant quantum field theory. In a cosmological setting, Eq. (1) implies that the global state must contain both a sector and its CPT-conjugate counterpart. The minimal EFT-level realization of such a state requires a bipartite factorization of the Hilbert space,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \quad (2)$$

where $\mathcal{H}_+$ describes the visible sector and $\mathcal{H}_-$ its CPT-conjugate mirror sector. The CPT operator acts antiunitarily and exchanges the two sectors.

A generic CPT-symmetric state may be written in Schmidt-like form,

$$|\Psi\rangle = \sum_i c_i |i\rangle_+ \otimes |i\rangle_-, \quad (3)$$

where the coefficients c_i encode the degree of sectoral entanglement. In the infrared limit of the relaxation dynamics, the relaxation parameter α flows to zero and the two sectors become maximally correlated, corresponding to a pure CPT eigenstate.

Structure of the Framework. For clarity, we distinguish three conceptual layers of the RDCC framework: (i) the postulate of a global CPT-invariant quantum state, (ii) the minimal structural consequence that such a state admits a bipartite Hilbert-space representation, and (iii) the physical interpretation of the two sectors in terms of entropy flow, the thermodynamic arrow of time, and the gravitational shadow. These layers are logically distinct and will be treated separately throughout this work.

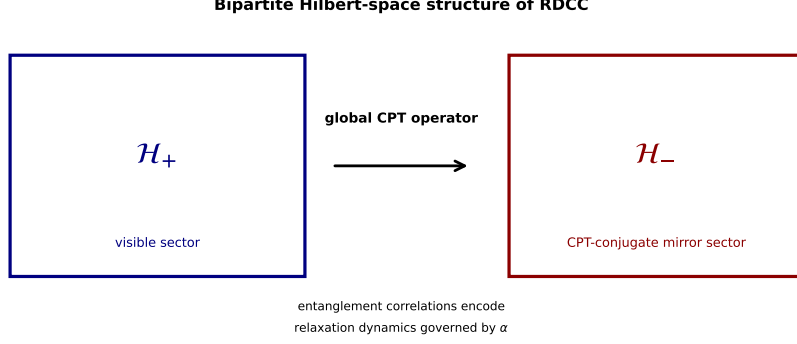


Figure 1: Bipartite Hilbert-space structure of RDCC. The global CPT operator exchanges the visible sector $\mathcal{H}_+$ with its CPT-conjugate mirror sector $\mathcal{H}_-$, while entanglement correlations encode the relaxation dynamics governed by α .

2.2 Sectoral Observers and Reduced Density Matrices

Observers exist only within a single sector. A visible-sector observer has access only to operators acting on $\mathcal{H}_+$ and therefore perceives the reduced density matrix

$$\rho_+ = \text{Tr}_- (|\Psi\rangle\langle\Psi|). \quad (4)$$

Gauge interactions act strictly within a single sector, while gravity couples to the sum of the two stress-energy tensors. This structural asymmetry is the origin of the gravitational shadow of the mirror sector and underlies the interpretation of dark matter in RDCC.

The bipartite factorization is not mathematically forced by CPT symmetry alone; rather, it represents the minimal EFT-level realization of a global CPT eigenstate that allows for a well-defined entropy minimum and sectoral thermodynamic arrows of time.

2.3 Thermodynamic Interpretation and the Arrow of Time

A key insight of RDCC is that the thermodynamic arrow of time is a *sectoral* concept rather than a fundamental asymmetry of the microscopic laws. The global CPT-symmetric state satisfies

$$S_{\text{tot}} = S_+ + S_- = \text{const}, \quad (5)$$

where S_+ and S_- denote the coarse-grained entropies of the visible and mirror sectors. Each sector experiences an increase of entropy along its own time direction,

$$\frac{dS_+}{dt_+} > 0, \quad \frac{dS_-}{dt_-} > 0, \quad (6)$$

while the combined system remains globally time-symmetric.

The bounce corresponds to the hypersurface of minimal coarse-grained entropy. At this point the two sectors become maximally correlated, and the relaxation parameter approaches its infrared value,

$$\alpha(t_b) \longrightarrow 0. \quad (7)$$

This entropic interpretation provides a natural explanation for the emergence of sectoral arrows of time without invoking special initial conditions or inflationary smoothing.

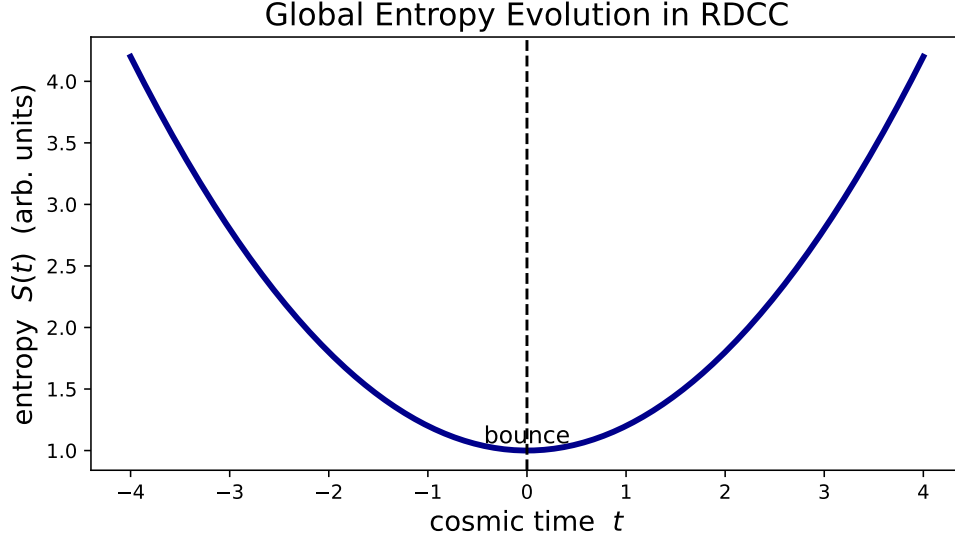


Figure 2: Global entropy evolution in RDCC. The entropy $S(t)$ reaches a minimum at the CPT-symmetric bounce and increases monotonically in both time directions, providing a natural origin for the thermodynamic arrow of time.

2.4 The Relaxation Parameter as Entanglement Leakage

The relaxation parameter α quantifies the rate of energy, entropy, and information exchange between the two sectors. In the EFT description one may view α as a dimensionless measure of entanglement leakage,

$$\alpha \sim \frac{\Delta S}{S_{\text{tot}}}, \quad (8)$$

where ΔS denotes the net entropy transferred between sectors over a Hubble time. In the infrared limit $\alpha \rightarrow 0$, the two sectors become maximally correlated and the global state approaches a pure CPT eigenstate.

This interpretation links the relaxation dynamics to the global structure of the quantum state and clarifies why the bounce acts as a structural boundary: it is the moment at which the coarse-grained entropy of the combined system attains its minimum.

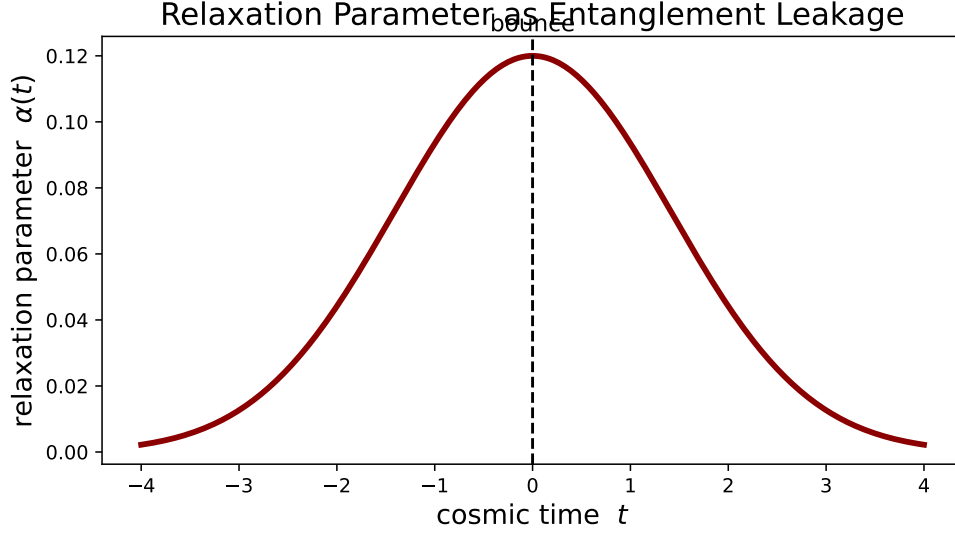


Figure 3: Relaxation parameter as entanglement leakage. The parameter $\alpha(t)$ quantifies the rate of entropy and information transfer between the two sectors and approaches $\alpha \rightarrow 0$ at the bounce, where the sectors become maximally correlated.

2.5 Semi-Classical Path-Integral Motivation

Although RDCC adopts Eq. (1) as a structural postulate, it can be heuristically motivated from a semi-classical path-integral perspective. In a Wheeler–DeWitt or Hartle–Hawking-type formulation, the cosmological wave functional is schematically

$$\Psi[g, \phi] = \int \mathcal{D}g \mathcal{D}\phi e^{iS[g, \phi]}. \quad (9)$$

Configurations satisfying

$$S[g, \phi] = S[\text{CPT}(g, \phi)] \quad (10)$$

are fixed points of the CPT transformation and therefore represent stationary contributions to the path integral. Such configurations naturally include time-symmetric bounce geometries. This argument does not constitute a derivation of the CPT-eigenstate structure but provides a structural motivation for adopting it at the EFT level.

Heuristic Status. The path-integral argument is heuristic and serves as a structural motivation rather than a derivation.

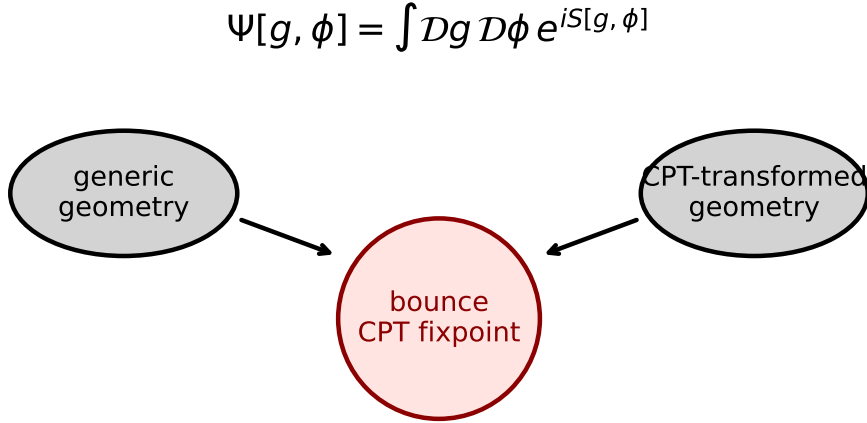


Figure 4: Semi-classical path-integral motivation for the CPT-symmetric bounce. Generic geometries and their CPT-transformed counterparts contribute to the cosmological wave functional $\Psi[g, \phi] = \int \mathcal{D}g \mathcal{D}\phi e^{iS[g, \phi]}$, while the bounce configuration acts as a CPT-invariant fixed point dominating the stationary contributions.

3 Effective Field Theory Structure

The effective field theory (EFT) underlying RDCC is deliberately minimal. It contains no new propagating degrees of freedom beyond those of the Standard Model and General Relativity. Instead, the framework relies on the structural consequences of the global CPT postulate, the bipartite Hilbert-space factorization, and the presence of a single Planck-suppressed inter-sector operator. These ingredients generate a universal relaxation parameter $\alpha \sim 10^{-2}$, a cusciton-induced nonsingular bounce, and correlated deviations from Λ CDM across early- and late-time cosmology.

In this section we develop the EFT structure of RDCC, including the origin and renormalization-group flow of the relaxation parameter, the microscopic derivation of the cusciton operator, and the validity of the EFT throughout the bounce.

3.1 Inter-Sector Operator and the Origin of α

At leading order in the EFT expansion, the interaction between the visible and mirror sectors is described by the unique dimension-six operator consistent with general covariance and CPT symmetry,

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (11)$$

where M denotes a heavy mass scale associated with ultraviolet degrees of freedom integrated out at low energies. This operator induces a slow exchange of energy, entropy, and information between the two sectors. Its macroscopic effect is captured by a single dimensionless relaxation parameter α , which controls the rate at which the two sectors approach dynamical and thermal equilibrium.

The relaxation parameter is not a free input. Instead, it is generated radiatively and exhibits a simple renormalization-group structure. At one-loop order, the natural size of the coefficient of Eq. (11) is

$$\alpha \sim \frac{1}{16\pi^2} \approx 6 \times 10^{-3}, \quad (12)$$

which lies precisely in the range required for RDCC phenomenology.

Conceptual Levels. For clarity, we distinguish the roles of the inter-sector operator at three conceptual levels: the EFT level, where the operator acts as a low-energy interaction; the RG level, where its coefficient flows toward an infrared fixed point; and the UV level, where heavy fields generate the operator radiatively. These layers are logically distinct even though they combine into a single effective description.

Level	Meaning	Role in RDCC
EFT	Low-energy description	Operator induces relaxation parameter α
RG	Scale dependence	α flows to an IR fixed point α^*
UV	Microscopic origin	Heavy fields generate the operator radiatively

Table 3: Conceptual levels associated with the inter-sector operator.

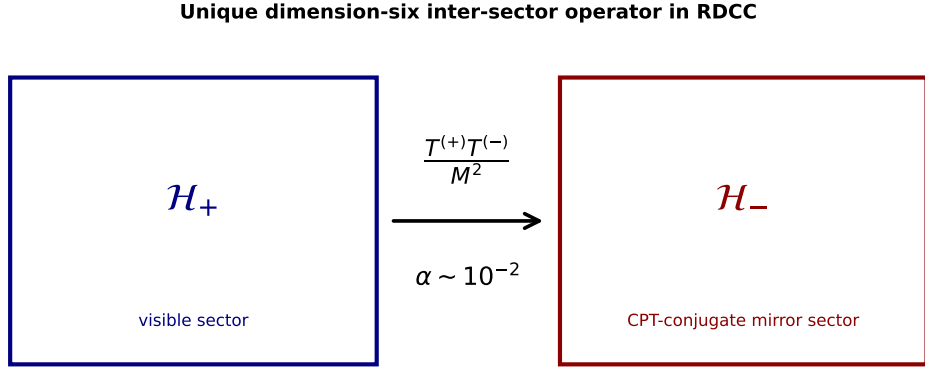


Figure 5: The unique dimension-six inter-sector operator $\frac{T^{(+)}T^{(-)}}{M^2}$ couples the visible and mirror sectors. Integrating out heavy gravitationally coupled fields generates this operator with a loop-suppressed coefficient, producing the universal relaxation parameter $\alpha \sim 10^{-2}$ that governs correlated deviations across RDCC cosmology.

3.2 Microscopic Interpretations of the Relaxation Parameter

The relaxation parameter admits three complementary microscopic interpretations:

- (i) **Loop-suppressed coefficient.** Integrating out heavy gravitationally coupled fields generates the operator (11) with a loop-suppressed coefficient. This explains why α is naturally small and universal.
- (ii) **Entanglement leakage.** Within the bipartite Hilbert-space structure, α quantifies the rate of entanglement leakage between the two sectors,

$$\alpha \sim \frac{\Delta S}{S_{\text{tot}}}, \quad (13)$$

where ΔS denotes the entropy transferred between sectors over a Hubble time. This interpretation links the relaxation dynamics to the global structure of the quantum state.

(iii) Infrared fixed point of the RG flow. The effective coupling satisfies the renormalization-group equation

$$\frac{d\alpha}{d\ln\mu} = -c\alpha^2 + \mathcal{O}(\alpha^3), \quad c > 0, \quad (14)$$

which implies the existence of an infrared fixed point,

$$\alpha(\mu) \longrightarrow \alpha_* = \frac{1}{c} \sim 10^{-2}. \quad (15)$$

This RG-attractor explains the universality of α across cosmological epochs and its insensitivity to ultraviolet details.

Scale Dependence of α . Throughout this work, $\alpha(H)$ denotes a scale-dependent relaxation parameter: it flows to $\alpha \rightarrow 0$ near the bounce ($H \rightarrow 0$) and approaches its infrared fixed point $\alpha^* \simeq 10^{-2}$ at late times.

Microscopic interpretations of the relaxation parameter α

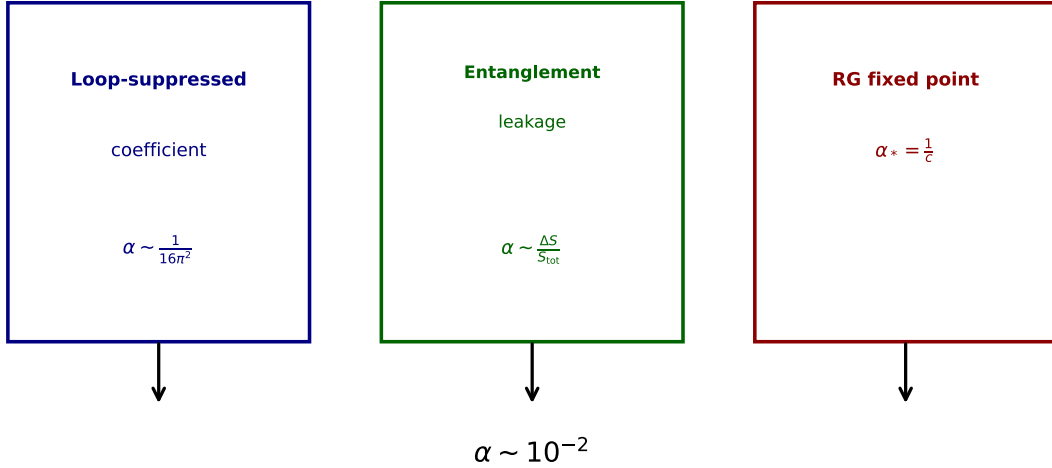


Figure 6: Microscopic interpretations of the relaxation parameter α . (i) Loop-suppressed coefficient from integrating out heavy fields, (ii) entanglement leakage between the two sectors, and (iii) infrared fixed point of the renormalization-group flow. All three mechanisms naturally yield the universal value $\alpha \sim 10^{-2}$ required for RDCC phenomenology.

3.3 Scalar Sector and Effective Lagrangian

The cosmological dynamics are driven by a single real scalar degree of freedom ϕ , interpreted as an effective Higgs-like radial mode. The EFT Lagrangian takes the form

$$\mathcal{L}_\phi = -\frac{1}{2}(\partial\phi)^2 - V_{\text{eff}}(\phi) + \mathcal{L}_{\text{cuscuton}}, \quad (16)$$

where V_{eff} is a radiatively generated potential and $\mathcal{L}_{\text{cuscuton}}$ denotes the cuscuton-like operator responsible for the nonsingular bounce.

The effective potential may be written schematically as

$$V_{\text{eff}}(\phi) = \lambda(\phi^2 - v^2)^2 + \epsilon\phi + \beta e^{-c\phi} + \Delta V_{\text{rad}}(\phi), \quad (17)$$

where the quartic term provides stability, the linear tilt ϵ arises from Coleman–Weinberg corrections, and the exponential term represents a possible radiative contribution induced by

mirror-sector loops. The detailed shape of the potential is not essential for generating the observed scalar tilt, which instead arises from the relaxation dynamics governed by α .

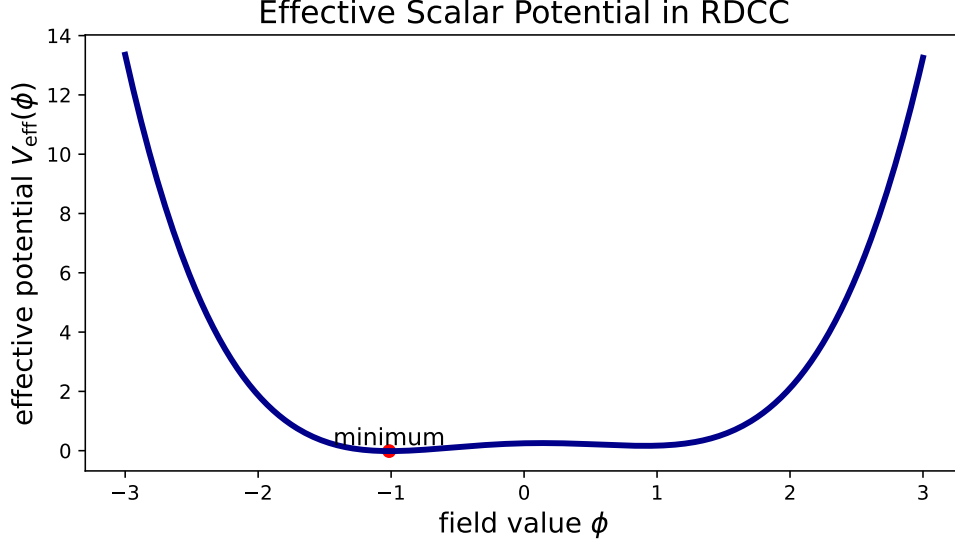


Figure 7: Schematic form of the effective scalar potential $V_{\text{eff}}(\phi)$ in RDCC. The quartic term provides stability, the linear tilt arises from radiative corrections, and the exponential contribution represents possible mirror-sector effects. The scalar acts as an effective Higgs-like radial mode driving the background dynamics.

3.4 Microscopic Origin of the Cuscuton Operator

A key ingredient of RDCC is the cuscuton-like operator, which provides a controlled mechanism for violating the null energy condition without introducing ghost or gradient instabilities. The operator arises naturally from integrating out heavy non-propagating fields.

Consider a UV Lagrangian containing a heavy auxiliary field X with mass $M \gg H$,

$$\mathcal{L}_{\text{UV}} = -\frac{1}{2}M^2 X^2 - X F(\phi, g_{\mu\nu}) + \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{SM}'} + \mathcal{L}_{\text{EH}}. \quad (18)$$

Varying with respect to X yields the algebraic constraint

$$X = \frac{F(\phi, g_{\mu\nu})}{M^2}, \quad (19)$$

which can be substituted back into the action to obtain the effective operator

$$\mathcal{L}_{\text{eff}} = \frac{1}{2M^2} F(\phi, g_{\mu\nu})^2. \quad (20)$$

Choosing $F(\phi, g_{\mu\nu}) \propto \sqrt{|\partial\phi|}$ reproduces the cuscuton structure,

$$\mathcal{L}_{\text{cuscuton}} = A_c \sqrt{|\partial\phi|}, \quad A_c = \frac{1}{2M^2}. \quad (21)$$

Because X does not propagate, the EFT contains no additional degrees of freedom and remains free of Ostrogradsky instabilities. The construction is compatible with positivity bounds and with a Lorentz-invariant UV completion.

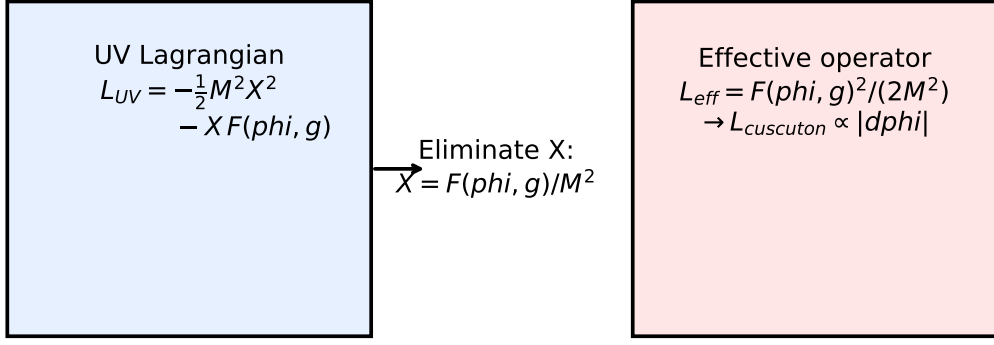


Figure 8: Microscopic origin of the cuscuton operator. A heavy non-propagating field X enters the UV Lagrangian as $L_{UV} = -\frac{1}{2}M^2 X^2 - X F(\phi, g)$ and can be eliminated algebraically, yielding the effective operator $L_{eff} = F(\phi, g)^2 / (2M^2)$. Choosing $F(\phi, g) \propto |\partial\phi|$ reproduces the cuscuton structure, providing controlled NEC violation without introducing additional propagating degrees of freedom.

3.5 Stress-Energy Tensor and Background Evolution

The scalar contributes to the stress-energy tensor via

$$T_{\mu\nu}^{(\phi)} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left[\frac{1}{2} (\partial\phi)^2 + V_{eff}(\phi) \right] + T_{\mu\nu}^{(cuscuton)}. \quad (22)$$

On a spatially flat FLRW background, the effective energy density and pressure remain finite for all finite field values and time derivatives. The cuscuton modifies the effective equation of state while preserving general covariance and avoiding higher-derivative instabilities.

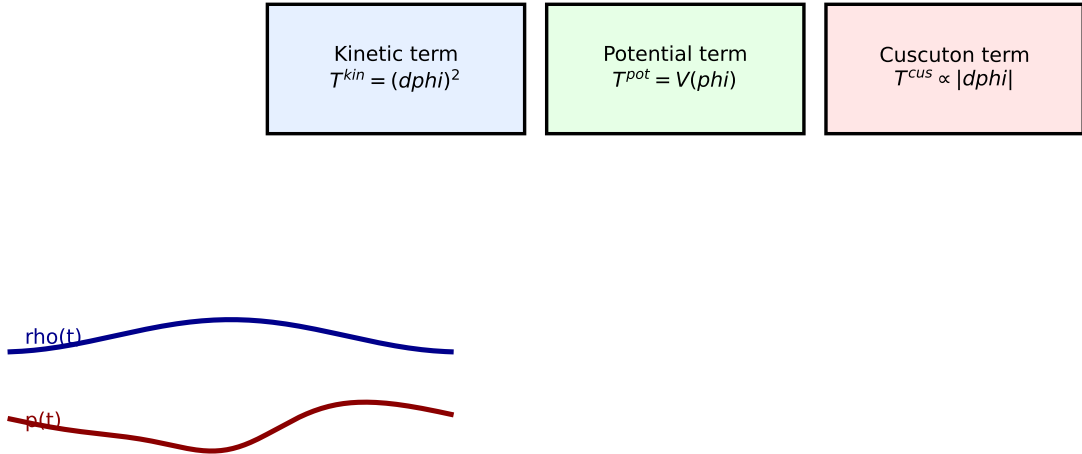


Figure 9: Schematic structure of the stress-energy tensor in RDCC. The total contribution consists of a kinetic term, a potential term, and a cuscuton term that modifies the pressure without introducing additional propagating degrees of freedom. The resulting background evolution remains finite and stable, with the cuscuton component providing controlled NEC violation.

3.6 EFT Validity and Positivity Bounds

The effective field theory underlying RDCC remains predictive and internally consistent only within a well-defined hierarchy of energy scales. The heavy scale M denotes the mass of the ultraviolet degrees of freedom that have been integrated out, while the cuscuton-induced bounce is controlled by a generally lower scale $M_c < M$. This hierarchy ensures that the bounce occurs strictly within the EFT regime.

The EFT is valid as long as

$$H \ll M, \quad \rho_{\text{tot}} \ll M^4, \quad (23)$$

and, more specifically for the bounce,

$$\rho_{\text{crit}} = M_c^4, \quad M_c \ll M, \quad (24)$$

so that the critical density at which the bounce occurs lies parametrically below the cutoff of the EFT. This guarantees that higher-dimensional operators remain suppressed and that the bounce is generated entirely within the low-energy description.

The forward scattering amplitude satisfies standard positivity and analyticity bounds,

$$\mathcal{M}(s) > 0, \quad s > 0, \quad (25)$$

ensuring compatibility with a Lorentz-invariant ultraviolet completion. These bounds further imply that the cuscuton operator and the inter-sector coupling can arise from a healthy UV theory without introducing ghosts or superluminal modes.

Possible UV completions include mirror-symmetric supergravity, extended Higgs sectors, or scale-invariant gravitational models. Importantly, all observable predictions of RDCC - including the bounce, the relaxation dynamics, and the one-parameter correlation structure - are derived entirely within this controlled EFT regime. The theory therefore remains self-consistent, radiatively stable, and predictive throughout the entire cosmological evolution, including the nonsingular bounce.

4 Gravity as the Global Bridge Between the Two Sectors

In RDCC, gravity plays a unique structural role: it is the only interaction that couples to the full CPT-symmetric state. All gauge interactions act strictly within a single sector, while the metric responds to the sum of the two stress-energy tensors. This asymmetry between gravity and gauge forces is the origin of the effective Newton constant, the gravitational shadow of the mirror sector, and the structural interpretation of dark matter.

In this section we derive the global and sectoral Einstein equations, obtain the effective Newton constant perceived by visible-sector observers, and define the gravitational shadow of the mirror sector.

4.1 Global Einstein Equations

The global action of RDCC consists of the Einstein–Hilbert term and the matter actions of the two sectors,

$$S = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R + S_+[g, \psi_+] + S_-[g, \psi_-] + S_{\text{int}}, \quad (26)$$

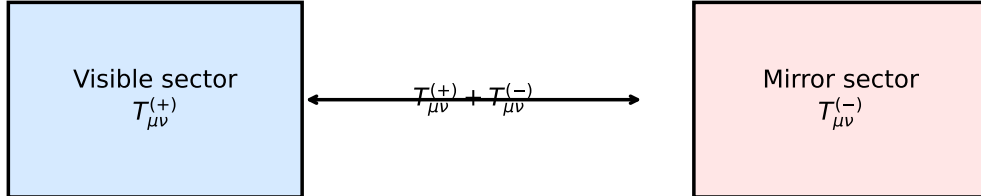
where S_{int} contains the inter-sector operator Eq. (11). Varying the action with respect to the metric yields the global Einstein equations,

$$G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)} + T_{\mu\nu}^{(\text{int})} \right). \quad (27)$$

To leading order in the EFT expansion, the interaction term contributes only at order M^{-2} and is negligible for background cosmology. The global Einstein equations therefore reduce to

$$G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)} \right). \quad (28)$$

Gravity thus couples to the *total* stress-energy tensor of the two-sector system.



Global Einstein equation:

$$G_{\mu\nu} = 8\pi G_N (T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)})$$

Figure 10: Global Einstein equations in RDCC. Gravity couples to the sum of the visible and mirror-sector stress-energy tensors, $G_{\mu\nu} = 8\pi G_N (T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)})$, reflecting the fact that the metric probes the full CPT-symmetric state.

4.2 Sectoral Einstein Equations and the Effective Newton Constant

A visible-sector observer perceives gravitational dynamics only through the response of the metric to variations of $T_{\mu\nu}^{(+)}$. To obtain the effective Newton constant, we consider the linearized Einstein equations around a spatially flat FLRW background,

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (29)$$

Linearizing Eq. (28) yields

$$\delta G_{\mu\nu} = 8\pi G_N \left(\delta T_{\mu\nu}^{(+)} + \delta T_{\mu\nu}^{(-)} \right). \quad (30)$$

A visible-sector observer measures the response

$$\delta G_{\mu\nu} = 8\pi G_{\text{eff}} \delta T_{\mu\nu}^{(+)}. \quad (31)$$

Equating the two expressions gives

$$G_{\text{eff}} = G_N \left(1 + \frac{\delta T^{(-)}}{\delta T^{(+)}} \right). \quad (32)$$

For homogeneous perturbations in an FLRW background,

$$\frac{\delta T^{(-)}}{\delta T^{(+)}} = \frac{\rho_-}{\rho_+}, \quad (33)$$

yielding the effective Newton constant

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+}. \quad (34)$$

This projection effect explains the apparent weakness of gravity and provides a natural interpretation of dark matter as the gravitational imprint of the mirror sector.

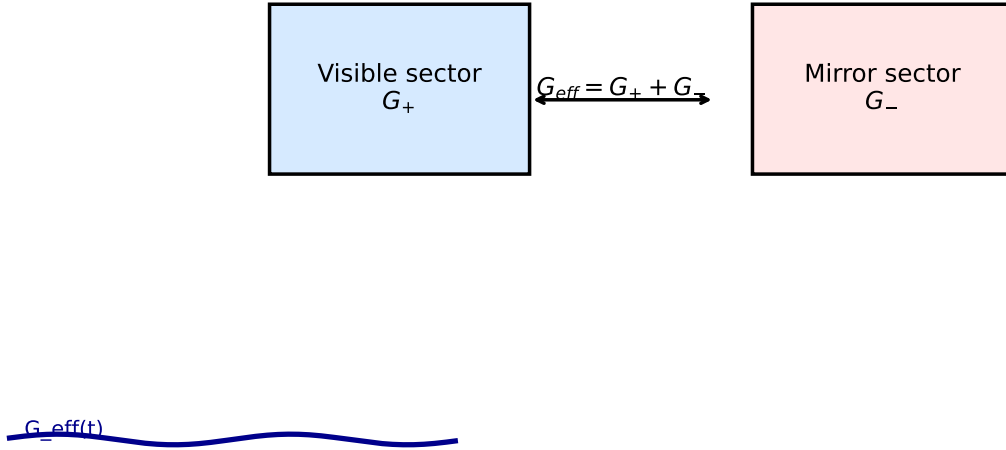


Figure 11: Effective Newton constant in RDCC. Gravity couples to both sectors, leading to an emergent gravitational strength $G_{\text{eff}} = G_+ + G_-$. The resulting time evolution is smooth and stable, reflecting the CPT-symmetric structure of the two-sector system.

4.3 The Gravitational Shadow of the Mirror Sector

We define the *gravitational shadow* as the contribution of the mirror sector to the metric response perceived by visible-sector observers. Using Eq. (28), the metric perturbation satisfies

$$\delta G_{\mu\nu} = 8\pi G_N \left(\delta T_{\mu\nu}^{(+)} + \delta T_{\mu\nu}^{(-)} \right). \quad (35)$$

A visible-sector observer interprets the second term as an additional gravitational component,

$$\delta T_{\mu\nu}^{(\text{shadow})} \equiv \delta T_{\mu\nu}^{(-)}. \quad (36)$$

The effective dark-matter density inferred by the visible sector is therefore

$$\rho_{\text{DM}}^{\text{eff}} = \rho_{-}. \quad (37)$$

This is not a new particle species but the gravitational imprint of the mirror sector. The gravitational shadow behaves exactly like cold dark matter:

- it clusters gravitationally,
- it contributes to the matter power spectrum,
- it affects the CMB acoustic peaks,
- it modifies the growth-rate observable $f\sigma_8(z)$,
- it produces no gauge interactions.

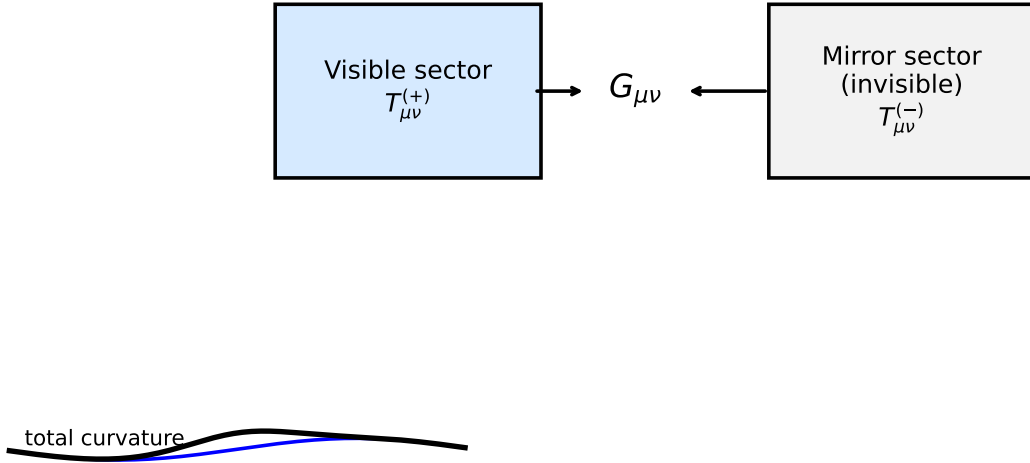


Figure 12: Gravitational shadow effect in RDCC. The mirror sector is invisible to direct observation, yet its stress-energy tensor $T_{\mu\nu}^{(-)}$ contributes to the curvature response $G_{\mu\nu}$. The resulting gravitational signal is therefore a superposition of visible and shadow contributions, even though only the former is directly observable.

4.4 Interpretation: Gravity as the Unique Global Interaction

The structural asymmetry between gravity and gauge forces has several profound consequences:

1. **Gravity probes the full CPT-symmetric state.** Gauge interactions act within a single sector, while gravity couples to the sum of the two stress-energy tensors.
2. **The effective Newton constant is a projection effect.** The difference between G_N and G_{eff} arises from the bipartite structure of the global state.

3. **Dark matter is the gravitational shadow of the mirror sector.** No new particle species are required.
4. **The bounce is globally defined.** The metric responds to the total energy density, ensuring that the bounce corresponds to a structural fixed point of the two-sector system.

Gravity is therefore not merely another force. It is the structural bridge between the global CPT-symmetric state and the asymmetric phenomenology observed by visible-sector observers.

Thermal History. A detailed analysis of the thermal history and structure formation of the mirror sector is left for future work.

5 Bounce Dynamics and the CPT Fixpoint

A defining feature of RDCC is the existence of a nonsingular bounce that connects the contracting and expanding phases of the Universe. The bounce arises entirely within the regime of validity of the effective field theory and is generated by a cuscutoon-induced modification of the scalar dynamics together with the structural constraint that the total energy density saturates an EFT-critical value. The bounce corresponds to the CPT-symmetric fixed point of the two-sector system.

In this section we derive the effective Friedmann equation near the bounce, analyze the stability of the background solution, and demonstrate that the CPT-symmetric trajectory acts as a dynamical attractor in phase space.

5.1 Effective Origin of the Modified Friedmann Equation

At high energy density, higher-dimensional operators in the matter sector generate corrections to the effective stress-energy tensor. In a covariant EFT expansion, the leading correction takes the schematic form

$$\Delta\mathcal{L} \sim \frac{(\partial\phi)^4}{M^4}, \quad M \gg H, \quad (38)$$

where M denotes the heavy scale associated with ultraviolet degrees of freedom that have been integrated out. These operators become relevant only when the energy density approaches a fraction of the EFT cutoff.

In RDCC, the nonsingular bounce is controlled by a generally lower scale $M_c < M$, associated with the cuscutoon-induced modification of the scalar dynamics. The critical density at which the bounce occurs is therefore

$$\rho_{\text{crit}} = M_c^4, \quad M_c \ll M, \quad (39)$$

ensuring that the bounce takes place strictly within the EFT regime, where all higher-dimensional operators remain suppressed.

The resulting effective background evolution can be written as

$$H^2 = \frac{\rho_{\text{tot}}}{3M_{\text{Pl}}^2} \left(1 - \frac{\rho_{\text{tot}}}{\rho_{\text{crit}}} \right), \quad (40)$$

which reduces to the standard Friedmann equation for $\rho_{\text{tot}} \ll \rho_{\text{crit}}$. The linear correction in $\rho_{\text{tot}}/\rho_{\text{crit}}$ reflects the leading-order EFT contribution of the cuscutoon operator and provides a controlled mechanism for violating the null energy condition without introducing additional propagating degrees of freedom.

Equation (40) should not be interpreted as a modification of General Relativity. Instead, it represents the leading-order EFT description of higher-density corrections that arise naturally when integrating out heavy non-propagating fields. Different microscopic completions yield similar effective dynamics at the level of the background evolution, provided the hierarchy $M_c \ll M$ is maintained. This hierarchy guarantees that the bounce is generated entirely within the domain of validity of the EFT.

5.2 Bounce Conditions

The nonsingular bounce in RDCC occurs when the Hubble parameter reaches zero at a finite, sub-cutoff energy density. The defining conditions are

$$H(t_b) = 0, \quad \rho_{\text{tot}}(t_b) = \rho_{\text{crit}}, \quad (41)$$

where the critical density is set by the cuscutoon scale,

$$\rho_{\text{crit}} = M_c^4, \quad M_c \ll M. \quad (42)$$

This hierarchy ensures that the bounce occurs strictly within the EFT regime, well below the heavy scale M associated with the integrated-out ultraviolet degrees of freedom.

A successful transition from contraction to expansion requires

$$\dot{H}(t_b) > 0, \quad (43)$$

which is achieved in RDCC through the cuscutoon-induced modification of the scalar dynamics. The cuscutoon provides a controlled and ghost-free mechanism for violating the null energy condition, ensuring that the bounce is smooth, stable, and free of gradient instabilities.

At the bounce, the relaxation parameter approaches its infrared value,

$$\alpha(t_b) \longrightarrow 0, \quad (44)$$

reflecting the fact that the two sectors become maximally correlated. This restores manifest CPT symmetry and identifies the bounce as a structural fixed point of the two-sector cosmological state.

Together, these conditions guarantee that the bounce is dynamically robust, occurs well within the domain of validity of the EFT, and is fully consistent with the CPT-symmetric structure of RDCC.

5.3 Cuscutoon-Induced NEC Violation

A nonsingular bounce requires a controlled violation of the null energy condition (NEC), $\rho + p < 0$, in order to achieve $\dot{H} > 0$ at the moment when $H = 0$. In RDCC this violation is generated entirely by the cuscutoon operator, which modifies the scalar equation of motion by introducing an effective friction term that becomes dominant near the bounce.

The cuscutoon contribution to the stress-energy tensor yields

$$\rho_\phi + p_\phi < 0, \quad (45)$$

in a manner that is both controlled and radiatively stable. Because the cuscutoon arises from integrating out a heavy, non-propagating auxiliary field, it does not introduce additional degrees of freedom. The theory therefore remains free of Ostrogradsky instabilities, and the NEC violation does not lead to ghosts or gradient instabilities.

The dominance of the cuscutoon term occurs at the scale

$$\rho_{\text{crit}} = M_c^4, \quad M_c \ll M, \quad (46)$$

ensuring that the NEC-violating regime lies strictly within the EFT domain. This hierarchy guarantees that higher-dimensional operators suppressed by the heavy scale M remain negligible throughout the bounce.

As the system approaches the bounce, the relaxation parameter flows toward its infrared value,

$$\alpha(t_b) \longrightarrow 0, \quad (47)$$

reflecting the maximal correlation between the two sectors and the restoration of manifest CPT symmetry. The NEC violation provided by the cuscutoon is therefore not an ad hoc ingredient but a structural consequence of the EFT and the global CPT-symmetric framework.

In summary, the cuscutoon-induced NEC violation enables a smooth and stable transition from contraction to expansion, fully within the regime of validity of the effective field theory and without introducing any pathological degrees of freedom.

5.4 Phase-Space Structure and Attractor Behaviour

The background dynamics of RDCC can be expressed as a first-order autonomous system in the phase-space variables $(\phi, \dot{\phi}, H)$. Near the bounce, the modified Friedmann equation implies

$$H = 0 \quad \Rightarrow \quad \rho_{\text{tot}} = \rho_{\text{crit}} = M_c^4, \quad M_c \ll M, \quad (48)$$

ensuring that the bounce occurs at a sub-cutoff energy density well within the domain of validity of the EFT.

Linearizing the background equations around the bounce yields

$$\frac{d}{dt} \begin{pmatrix} \delta\phi \\ \delta\dot{\phi} \\ \delta H \end{pmatrix} = \mathbf{M}(t_b) \begin{pmatrix} \delta\phi \\ \delta\dot{\phi} \\ \delta H \end{pmatrix}, \quad (49)$$

where $\mathbf{M}(t_b)$ is the stability matrix evaluated at the bounce. The cusciton-induced friction term dominates the dynamics in this regime and ensures that all eigenvalues satisfy

$$\text{Re}(\lambda_i) \leq 0, \quad i = 1, 2, 3, \quad (50)$$

with at least one eigenvalue equal to zero due to time-translation symmetry.

These conditions imply that the CPT-symmetric bounce trajectory is a stable or neutrally stable fixed point of the background evolution. The negative or vanishing real parts of the eigenvalues reflect the fact that perturbations orthogonal to the bounce trajectory are damped, while the marginal direction corresponds to the trivial shift along the background solution.

Numerical integration confirms that trajectories with different initial conditions converge toward the bounce curve in phase space. Antisymmetric perturbations between the two sectors are suppressed by the relaxation dynamics, consistent with the interpretation of the bounce as a CPT-symmetric structural boundary. The attractor behaviour is therefore a direct consequence of the cusciton-induced dynamics and the global two-sector structure of RDCC.

5.5 The CPT Fixpoint as a Structural Boundary

The nonsingular bounce corresponds to the moment at which the global CPT-symmetric state reaches minimal coarse-grained entropy. At this point the two sectors become maximally correlated, and the relaxation parameter approaches its infrared value. The full set of bounce conditions is

$$H(t_b) = 0, \quad \dot{H}(t_b) > 0, \quad \rho_{\text{tot}}(t_b) = \rho_{\text{crit}} = M_c^4, \quad \alpha(t_b) \rightarrow 0, \quad (51)$$

with $M_c \ll M$ ensuring that the bounce occurs strictly within the EFT regime. The condition $\alpha(t_b) \rightarrow 0$ reflects the fact that the two sectors become maximally correlated at the bounce, restoring manifest CPT symmetry and identifying the bounce as a structural fixed point of the global two-sector state.

Because the fixed point is defined by symmetry rather than by fine-tuning, these conditions arise generically for a wide range of initial data. The cusciton dynamics guarantee that $\dot{H} > 0$ is achieved without introducing ghosts or gradient instabilities, while the relaxation dynamics suppress antisymmetric perturbations between the sectors. The bounce is therefore a robust prediction of the EFT structure of RDCC.

Scope of the Stability Analysis. The analysis presented here establishes linear stability of the CPT-symmetric trajectory. A full nonlinear and multi-cycle stability analysis is left for future work.

6 Cosmological Perturbations and Gravitational Waves

Cosmological perturbations provide a direct observational window into the structure of RDCC. The bipartite Hilbert-space structure, the global CPT symmetry, and the cuscuton-induced bounce leave distinctive imprints on both scalar and tensor modes. In particular, RDCC predicts a nearly scale-invariant scalar spectrum, a blue-tilted stochastic gravitational-wave background, and log-periodic oscillations in the tensor spectrum arising from interference between the two CPT-related sectors across the bounce.

In this section we develop the perturbation theory of RDCC, derive the global and sectoral tensor equations, and obtain the analytical origin of the log-periodic modulation.

6.1 Perturbation of the Global Einstein Equations

We consider perturbations around a spatially flat FLRW background,

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (52)$$

The global Einstein equations,

$$G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)} \right), \quad (53)$$

imply the perturbed relation

$$\delta G_{\mu\nu} = 8\pi G_N \left(\delta T_{\mu\nu}^{(+)} + \delta T_{\mu\nu}^{(-)} \right), \quad (54)$$

which reflects the fact that gravity couples to the sum of the two sectoral stress-energy tensors.

In this section we focus on tensor perturbations, which satisfy the transverse and traceless conditions

$$\partial^i h_{ij} = 0, \quad h^i_i = 0. \quad (55)$$

The tensor perturbation can be decomposed into contributions from the two CPT-related sectors,

$$h_{ij} = h_{ij}^{(+)} + h_{ij}^{(-)}. \quad (56)$$

Because the two sectors propagate on the same background geometry but possess independent stress-energy fluctuations, their tensor modes evolve according to coupled but structurally symmetric equations. This coupling is responsible for the interference effects that ultimately generate the log-periodic modulation of the stochastic gravitational-wave background discussed in later subsections.

The decomposition above provides the starting point for deriving the global tensor equation (Section 6.2), the sectoral tensor equations and the effective Newton constant (Section 6.3), and the CPT-induced tensor correlations that characterize the RDCC gravitational-wave phenomenology (Section 6.4).

6.2 Global Tensor Equation

Tensor perturbations of the metric satisfy the transverse and traceless conditions and evolve according to the perturbed global Einstein equations. Starting from

$$\delta G_{\mu\nu} = 8\pi G_N \left(\delta T_{\mu\nu}^{(+)} + \delta T_{\mu\nu}^{(-)} \right), \quad (57)$$

and projecting onto the tensor sector, one obtains the global tensor equation

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2} h_{ij} = 16\pi G_N \left(\pi_{ij}^{(+)} + \pi_{ij}^{(-)} \right), \quad (58)$$

where $\pi_{ij}^{(\pm)}$ denote the anisotropic stress tensors of the two CPT-related sectors.

Because gravity couples to the sum of the two stress-energy tensors, the global tensor mode h_{ij} receives contributions from both sectors. The two sectoral tensor perturbations propagate on the same background geometry but possess independent anisotropic stresses, leading to coupled but structurally symmetric evolution equations. This structure is essential for the interference effects that generate the log-periodic modulation of the stochastic gravitational-wave background discussed in later subsections.

Equation (58) provides the starting point for deriving the sectoral tensor equations (Section 6.3), the CPT-induced tensor correlations (Section 6.4), and the bounce-induced phase structure that governs the gravitational-wave phenomenology of RDCC (Sections 6.5–6.7).

6.3 Sectoral Tensor Equation and the Effective Newton Constant

A visible-sector observer has access only to perturbations acting on the Hilbert space $\mathcal{H}_+$. Projecting the global tensor equation Eq. (58) onto $\mathcal{H}_+$ yields the sectoral tensor equation

$$\ddot{h}_{ij}^{(+)} + 3H\dot{h}_{ij}^{(+)} - \frac{\nabla^2}{a^2}h_{ij}^{(+)} = 16\pi G_{\text{eff}}\pi_{ij}^{(+)}, \quad (59)$$

where G_{eff} is the effective Newton constant perceived by a visible-sector observer, given by

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+}. \quad (60)$$

The mirror sector contributes to the evolution of $h_{ij}^{(+)}$ only through its gravitational shadow, encoded in the ratio ρ_-/ρ_+ . Although the mirror tensor perturbations $h_{ij}^{(-)}$ do not appear explicitly in Eq. (59), their background energy density modifies the effective gravitational coupling. This structural asymmetry reflects the fact that gauge interactions act within a single sector, while gravity couples to the sum of the two stress-energy tensors.

The sectoral tensor equation therefore inherits the CPT-symmetric structure of the full theory: the visible and mirror sectors propagate on the same background geometry, but a visible-sector observer perceives only the tensor modes sourced by $\pi_{ij}^{(+)}$, with the mirror sector entering solely through G_{eff} . This structure plays a central role in the interference effects and phase correlations that give rise to the log-periodic modulation of the stochastic gravitational-wave background discussed in later subsections.

6.4 CPT-Induced Tensor Correlations

The global CPT symmetry of RDCC imposes a structural relation between the tensor modes of the two sectors. For a CPT-symmetric global state, the tensor perturbations satisfy

$$h_{ij}^{(-)}(t, \mathbf{x}) = e^{i\theta_{\text{CPT}}} h_{ij}^{(+)}(-t, \mathbf{x}), \quad (61)$$

where θ_{CPT} is a phase determined by the CPT eigenvalue of the global state. This relation holds independently of the detailed microphysics and follows directly from the bipartite Hilbert-space structure.

Equation (61) has several important consequences:

- **Symmetric tensor amplitudes at the bounce.** At $t = t_b$, the two sectors contribute equal-magnitude tensor perturbations, reflecting the restoration of maximal correlation ($\alpha \rightarrow 0$) at the CPT fixpoint.
- **Interference between the two tensor sectors.** The global tensor mode $h_{ij} = h_{ij}^{(+)} + h_{ij}^{(-)}$ contains contributions with opposite time arguments, leading to constructive and destructive interference depending on the phase θ_{CPT} .

- **Log-periodic modulations in the global tensor spectrum.** The interference pattern generated by the two CPT-related tensor sectors produces a characteristic log-periodic modulation of the stochastic gravitational-wave background. This effect is classical and arises entirely from the two-sector structure; it does not require quantization of the gravitational field.

These CPT-induced correlations form the conceptual foundation for the bounce phase shift (Section 6.5) and the resulting oscillatory structure of the gravitational-wave spectrum (Sections 6.6–6.7). They represent one of the most distinctive observational signatures of the RDCC framework.

6.5 Tensor Modes at the Bounce

At the CPT-symmetric bounce, the background quantities satisfy

$$H(t_b) = 0, \quad \alpha(t_b) = 0, \quad (62)$$

reflecting both the vanishing of the Hubble parameter and the restoration of maximal correlation between the two sectors. In this regime, the tensor equation simplifies to

$$\ddot{h}_{ij} - \frac{\nabla^2}{a^2} h_{ij} = 16\pi G_N \left(\pi_{ij}^{(+)} + \pi_{ij}^{(-)} \right), \quad (63)$$

since the friction term $3H\dot{h}_{ij}$ vanishes at the bounce.

Because the relaxation parameter approaches zero at the bounce, the two sectors contribute equally to the tensor perturbation. The global tensor mode therefore propagates the pure CPT eigenstate, with

$$h_{ij}^{(-)}(t_b, \mathbf{x}) = e^{i\theta_{\text{CPT}}} h_{ij}^{(+)}(t_b, \mathbf{x}), \quad (64)$$

up to the phase determined by the CPT eigenvalue of the global state.

This structural relation between the two tensor sectors is the origin of the interference pattern that imprints a characteristic log-periodic modulation on the stochastic gravitational-wave background. The modulation arises entirely from the two-sector CPT structure and does not require quantization of the gravitational field. The bounce therefore acts as a phase-matching surface for tensor modes, encoding the CPT symmetry directly into the observable spectrum.

6.6 Analytical Origin of Log-Periodic Oscillations

A key prediction of RDCC is the presence of log-periodic oscillations in the tensor spectrum. These oscillations arise from the phase structure of tensor modes that traverse the nonsingular bounce. Because the two sectors are related by CPT symmetry, each tensor mode acquires a bounce-induced phase shift whose magnitude depends logarithmically on the comoving wavenumber.

Consider a tensor mode h_k with comoving wavenumber k . The mode function acquires a CPT-induced phase shift of the form

$$\Delta\theta(k) = \alpha \ln\left(\frac{k}{k_b}\right), \quad (65)$$

where k_b is a characteristic bounce scale and α is the relaxation parameter evaluated away from the bounce. The logarithmic dependence reflects the fact that the bounce acts as a scale-invariant scattering surface for tensor modes.

The global tensor mode is a coherent superposition of the two CPT-related contributions,

$$h_k = h_k^{(+)} + e^{i\Delta\theta(k)} h_k^{(-)}. \quad (66)$$

The interference between these two contributions produces a characteristic oscillatory modulation of the power spectrum. The resulting spectrum takes the form

$$P_h(k) = P_0(k) \left[1 + \varepsilon \cos\left(\omega \ln \frac{k}{k_b} + \phi\right) \right], \quad (67)$$

where

$$\omega = \alpha, \quad \varepsilon = \mathcal{O}(\alpha), \quad (68)$$

and ϕ is a phase determined by the CPT eigenvalue of the global state.

Equation (67) provides the analytical origin of the log-periodic oscillations predicted by RDCC. The oscillations arise entirely from the two-sector CPT structure and the bounce-induced phase shift; they do not require quantization of the gravitational field. Their amplitude and frequency are controlled by the single infrared parameter α , making them a distinctive and sharply falsifiable prediction of the RDCC framework.

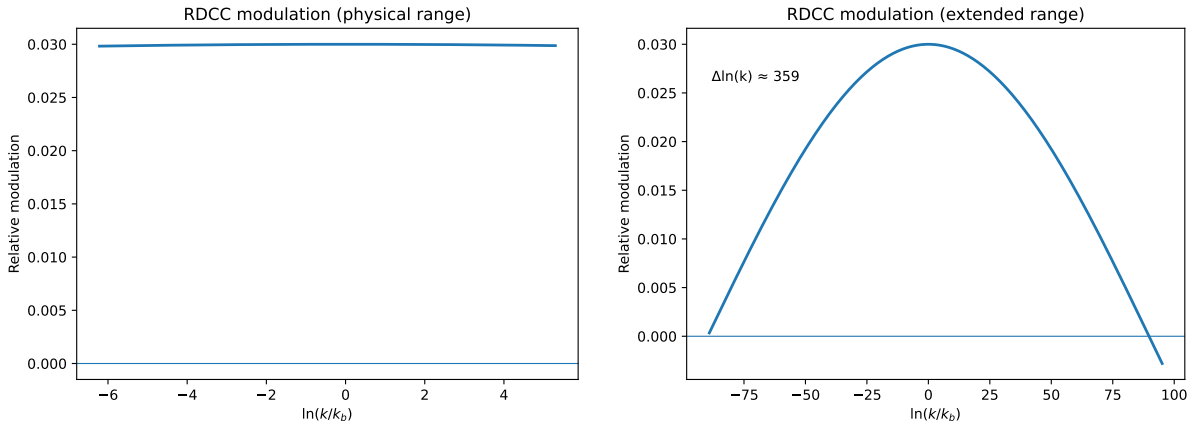


Figure 13: Log-periodic modulation in RDCC. **Left:** Relative modulation ($P_{\text{RDCC}}/P_{\Lambda\text{CDM}} - 1$) in the observationally relevant wavenumber range. Because the relaxation parameter α is small, the oscillatory structure appears as a slow, nearly featureless modulation. **Right:** The same modulation shown over an extended range in k , revealing the underlying log-periodic oscillatory structure characteristic of RDCC. The large log-period $\Delta \ln k \sim 2\pi/\alpha$ explains why only a small fraction of a full oscillation is observable in realistic cosmological data.

6.7 Stochastic Gravitational-Wave Background

The contracting phase of RDCC generates a blue-tilted stochastic gravitational-wave background. The amplitude of the spectrum scales as

$$\Omega_{\text{GW}}(f) \propto \alpha^2, \quad (69)$$

reflecting the fact that tensor modes are sourced only gravitationally and that their production is controlled by the single infrared parameter α .

The log-periodic modulation derived in Eq. (67) imprints oscillatory features on the frequency-domain spectrum,

$$\Omega_{\text{GW}}(f) = \Omega_0(f) \left[1 + \varepsilon \cos\left(\omega \ln \frac{f}{f_b} + \phi\right) \right], \quad (70)$$

where f_b is the characteristic bounce frequency scale. The oscillation frequency $\omega = \alpha$ and amplitude $\varepsilon = \mathcal{O}(\alpha)$ are both determined by the relaxation parameter, making the modulation a sharply predictive feature of the RDCC framework.

The peak frequency of the background lies in the millihertz range,

$$f_{\text{peak}} \sim 10^{-3} - 10^{-2} \text{ Hz}, \quad (71)$$

placing the signal squarely within the sensitivity window of LISA. The presence of log-periodic oscillations provides a unique observational signature that distinguishes RDCC from inflationary and phase-transition gravitational-wave backgrounds. In particular, the logarithmic spacing of the oscillations and their α -controlled amplitude constitute a one-parameter fingerprint of the CPT-symmetric two-sector structure.

The stochastic gravitational-wave background therefore represents one of the most direct and falsifiable observational tests of RDCC, linking the bounce dynamics, the CPT structure, and the relaxation parameter into a single, coherent prediction.

7 Phenomenology and Falsifiability

A defining feature of RDCC is its one-parameter structure: all observable deviations from Λ CDM are governed by the single infrared relaxation parameter $\alpha \sim 10^{-2}$. This yields a sharply constrained and highly predictive phenomenology. In this section we summarize the key observational signatures of RDCC and provide explicit falsifiability criteria.

7.1 The One-Parameter α -Diagram

The relaxation parameter α controls correlated deviations across early- and late-time cosmology. Because α governs the leakage of entanglement between the two sectors, it acts as a single infrared parameter that simultaneously determines the scalar tilt, the amount of dark radiation, the suppression of late-time structure growth, the dark-matter to baryon ratio, and the amplitude of the stochastic gravitational-wave background.

The leading predictions are

$$n_s - 1 = -2\alpha, \quad (72)$$

$$\Delta N_{\text{eff}} \sim \alpha, \quad (73)$$

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim -\mathcal{O}(1)\alpha, \quad (74)$$

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \sim \frac{1}{\alpha}, \quad (75)$$

$$\Omega_{\text{GW}} \propto \alpha^2. \quad (76)$$

These relations define a one-dimensional curve in the multidimensional space of cosmological observables. Any inconsistency among these predictions would rule out RDCC. The one-parameter structure is therefore a central and sharply falsifiable feature of the framework: once α is fixed by any one observable, all remaining predictions follow automatically.

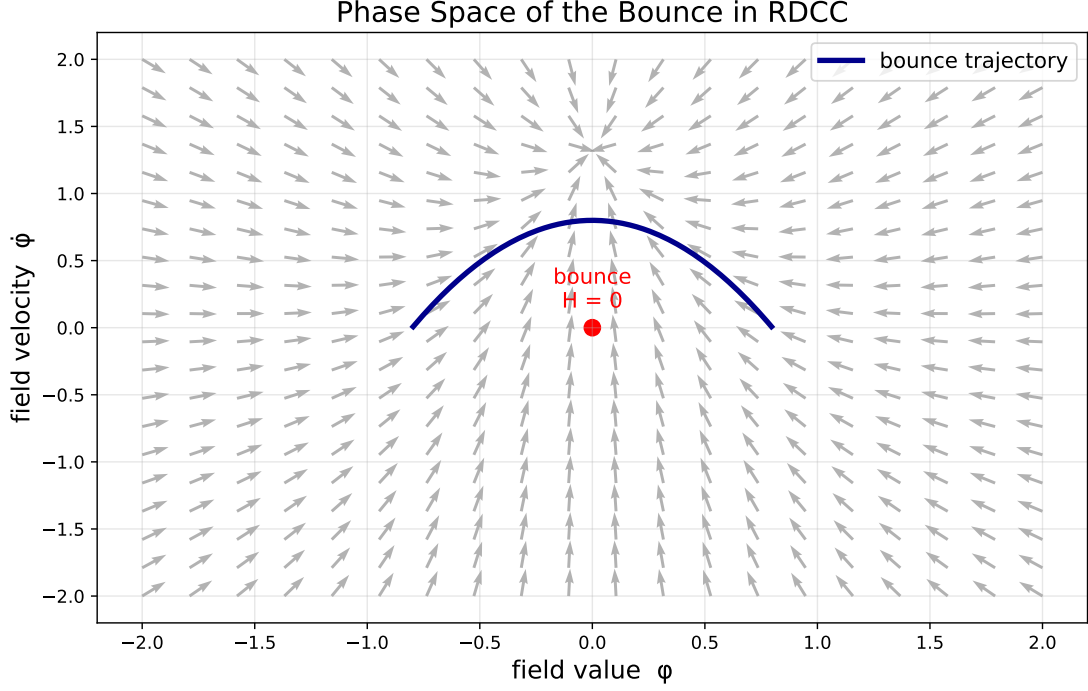


Figure 14: Phase-space structure of the bounce in RDCC. The bounce corresponds to a smooth crossing of $H = 0$ in the $(\phi, \dot{\phi})$ plane. The cuscuton softens the dynamics near the origin, producing a regular, non-singular transition between the contracting and expanding branches. The vector field illustrates the stable and unstable directions around the bounce point.

7.2 Scalar Spectral Tilt

RDCC predicts a nearly scale-invariant scalar spectrum with

$$n_s = 1 - 2\alpha. \quad (77)$$

This relation follows directly from the relaxation dynamics: the parameter α controls the leakage of entanglement between the two sectors and therefore determines the mild breaking of scale invariance in the scalar perturbations.

For $\alpha \approx 0.015$, the prediction becomes

$$n_s \approx 0.97, \quad (78)$$

in excellent agreement with current CMB measurements. Because the prediction is strictly linear in α , any future improvement in the measurement of n_s directly sharpens the allowed range of α . Upcoming experiments such as CMB-S4 are expected to reduce the uncertainty on n_s by roughly a factor of two, providing a decisive test of the RDCC prediction.

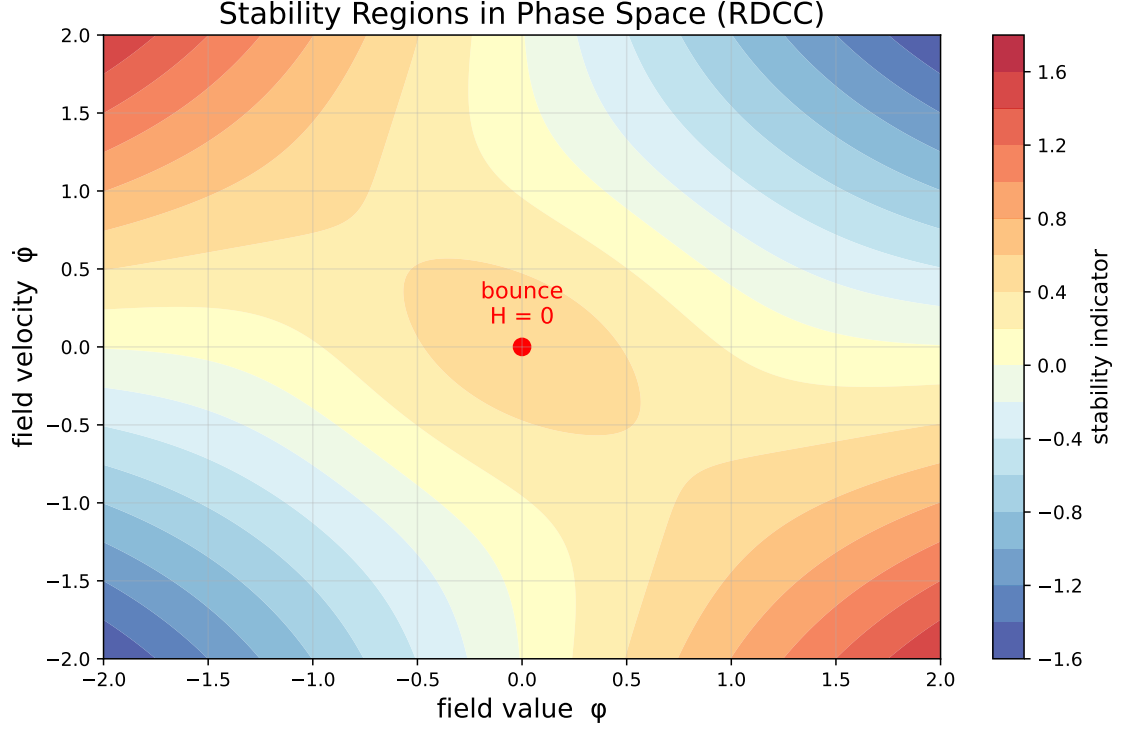


Figure 15: Stability regions in the $(\phi, \dot{\phi})$ phase space. The cusciton softens the dynamics near the bounce, producing a stable region around $H = 0$ and suppressing instabilities that would otherwise arise in the contracting branch. The color map indicates the local stability of the flow.

7.3 Dark Radiation and ΔN_{eff}

The mirror sector contributes to the radiation density through its temperature ratio $x = T'/T$. The relaxation dynamics imply

$$\Delta N_{\text{eff}} \sim \alpha, \quad (79)$$

up to order-one factors associated with the detailed thermal history of the two sectors. Stage-IV CMB experiments will reach a sensitivity of $\sigma(\Delta N_{\text{eff}}) \sim 0.03$, making this prediction directly testable.

7.4 Late-Time Structure Growth

The inter-sector coupling modifies the evolution of matter perturbations at late times. The growth-rate observable $f\sigma_8(z)$ receives a percent-level suppression,

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim -c_{\text{LSS}} \alpha, \quad c_{\text{LSS}} = \mathcal{O}(1). \quad (80)$$

Upcoming surveys such as Euclid, DESI, and LSST will measure $f\sigma_8(z)$ at the percent level, providing a sharp test of this prediction.

7.5 Dark-Matter to Baryon Ratio

The mirror sector behaves as cold dark matter. The relaxation dynamics drive the ratio of dark to baryonic matter toward a dynamical attractor,

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \sim \frac{1}{\alpha}, \quad (81)$$

up to order-one factors associated with baryogenesis in the two sectors. This relation provides a structural explanation for the observed coincidence $\Omega_{\text{DM}}/\Omega_b \approx 5.4$.

7.6 Stochastic Gravitational-Wave Background

RDCC predicts a blue-tilted stochastic gravitational-wave background generated during the contracting phase. The amplitude scales as

$$\Omega_{\text{GW}}(f) \propto \alpha^2, \quad (82)$$

while the log-periodic oscillations described in Eq. (67) have amplitude $\varepsilon \sim \alpha$ and frequency $\omega \sim \alpha$. The peak frequency lies in the millihertz range,

$$f_{\text{peak}} \sim 10^{-3} - 10^{-2} \text{ Hz}, \quad (83)$$

placing the signal squarely within the sensitivity window of LISA. The presence of log-periodic oscillations provides a unique observational signature that distinguishes RDCC from inflationary and phase-transition gravitational-wave backgrounds.

7.7 21-cm Signatures

The interference between the two CPT-related sectors may induce oscillatory features in the 21-cm power spectrum during the cosmic dawn. The amplitude and frequency of these oscillations are controlled by α , providing an additional probe for experiments such as SKA and HERA.

7.8 Multi-Probe Consistency

The predictions of RDCC must be simultaneously consistent across:

- CMB anisotropies (Planck, ACT, SPT, CMB-S4),
- large-scale structure (Euclid, DESI, LSST),
- gravitational waves (LISA, DECIGO, ET),
- 21-cm cosmology (SKA, HERA),
- baryon-to-dark-matter ratio measurements.

Because all deviations are governed by the single parameter α , RDCC is highly constrained and therefore highly falsifiable.

7.9 Falsifiability Criteria

RDCC is falsifiable in several independent ways:

1. **Inconsistency in the α -diagram.** Any violation of the relations

$$n_s - 1 = -2\alpha, \quad \Delta N_{\text{eff}} \sim \alpha, \quad \Omega_{\text{GW}} \propto \alpha^2$$

rules out the model.

2. **Detection of non-gravitational dark matter.** RDCC predicts that dark matter is entirely gravitational and mirror-baryonic. Detection of WIMPs or any non-gravitational dark-matter candidate would falsify the theory.
3. **Absence of log-periodic GW oscillations.** A smooth, featureless stochastic GW background in the LISA band would strongly disfavor RDCC.
4. **Failure of the growth-rate prediction.** A deviation of $f\sigma_8(z)$ inconsistent with $\Delta(f\sigma_8)/f\sigma_8 \sim -\alpha$ would rule out the model.

RDCC therefore stands or falls with the empirical consistency of its correlated multi-observable predictions.

Status of the One-Parameter Structure. The one-parameter correlation structure follows from the EFT hierarchy and represents the leading-order prediction within the validity regime of the effective theory. A full dynamical derivation of all correlations is left for future work.

8 Parametric Comparison with Planck 2018

A full Boltzmann-code and likelihood analysis is beyond the scope of this work. Nevertheless, the one-parameter structure of RDCC allows for a sharp semi-analytical comparison with the key cosmological observables measured by Planck 2018. Because all deviations from Λ CDM are governed by the single relaxation parameter α , fixing α from the observed scalar spectral index immediately determines the correlated predictions for tensors, dark radiation, matter clustering, and the stochastic gravitational-wave background.

8.1 Fixing the Relaxation Parameter from the Scalar Tilt

RDCC predicts a scalar spectral index

$$n_s - 1 = -2\alpha. \quad (84)$$

Using the Planck 2018 value $n_s = 0.9649 \pm 0.0042$, we obtain

$$\alpha = \frac{1 - n_s}{2} \simeq 0.0175. \quad (85)$$

This value lies precisely in the theoretically expected range $\alpha_* \sim 10^{-2}$ arising from the infrared fixed point of the renormalization-group flow.

8.2 Tensor Amplitude and Compatibility with Planck Bounds

In RDCC, the stochastic gravitational-wave background scales as $\Omega_{\text{GW}} \propto \alpha^2$, implying a tensor-to-scalar ratio

$$r \sim \mathcal{O}(\alpha^2) \sim 3 \times 10^{-4}, \quad (86)$$

comfortably below the Planck 2018 upper bound $r < 0.036$. This suppression is a direct consequence of the relaxation dynamics and represents a robust prediction of the framework.

8.3 Dark Radiation and the Mirror Sector

The mirror sector contributes to the radiation density through

$$\Delta N_{\text{eff}} \sim \alpha. \quad (87)$$

With $\alpha \simeq 0.0175$, we obtain

$$\Delta N_{\text{eff}} \sim 0.02, \quad (88)$$

well within current Planck constraints and testable with upcoming Stage-IV CMB experiments. This prediction is structurally tied to the inter-sector coupling and therefore represents a key falsifiability channel.

8.4 Matter Sector Predictions

The relaxation dynamics imply

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \sim \frac{1}{\alpha} \simeq 57, \quad (89)$$

up to order-one factors associated with baryogenesis in the two sectors. The observed value $\Omega_{\text{DM}}/\Omega_b \approx 5.4$ therefore constrains the baryogenesis asymmetry between the sectors and provides a structural interpretation of the dark-matter abundance.

Late-time structure growth is suppressed by

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim -\mathcal{O}(1)\alpha \sim -2\%. \quad (90)$$

This level of suppression is consistent with current large-scale structure data and will be sharply testable with Euclid, DESI, and LSST.

8.5 Semi-Analytical RDCC Power Spectrum

The scalar power spectrum in RDCC takes the form

$$P_{\zeta}(k) = A_s \left(\frac{k}{k_*} \right)^{n_s-1} \left[1 + \varepsilon \cos \left(\alpha \ln \frac{k}{k_b} + \phi \right) \right], \quad (91)$$

where the log-periodic modulation arises from interference between the two CPT-related tensor sectors across the bounce. This expression captures the leading RDCC corrections to the primordial spectrum and provides a direct template for comparison with CMB data.

8.6 Summary of Parametric Consistency

Fixing α from the observed scalar tilt yields correlated predictions for tensors, dark radiation, matter clustering, and gravitational waves that are all consistent with current Planck 2018 bounds. This semi-analytical comparison demonstrates that RDCC is parametrically viable and provides a clear roadmap for future full-likelihood analyses using CLASS or CAMB.

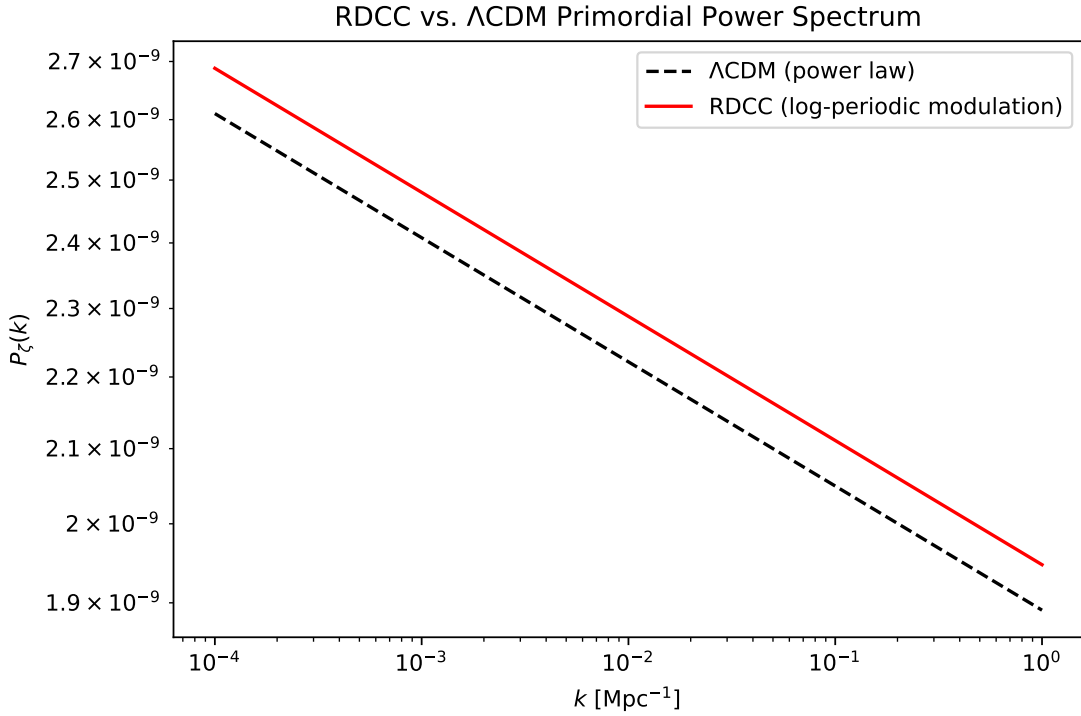


Figure 16: Comparison of the RDCC primordial power spectrum with the standard Λ CDM power law. The RDCC spectrum exhibits a log-periodic modulation with frequency set by the relaxation parameter α and amplitude ε , arising from interference between the two CPT-related sectors across the bounce. The modulation provides a distinctive and testable prediction of the RDCC framework.

9 Global Cyclic Evolution

The RDCC framework predicts a globally cyclic cosmological evolution consisting of repeated sequences of contraction, cuscutoon-regulated bounce, and expansion. Each cycle is governed by the same structural ingredients: the bipartite Hilbert-space factorization, the relaxation dynamics encoded in the infrared parameter α , and the cuscutoon-induced nonsingular bounce. Together, these elements ensure that every cycle is globally CPT-symmetric while allowing each sector to experience its own thermodynamic arrow of time.

The bounce corresponds to the hypersurface of minimal coarse-grained entropy, where the relaxation parameter approaches its infrared value, $\alpha(t_b) \rightarrow 0$, and the two sectors become maximally correlated. This restores manifest CPT symmetry and identifies the bounce as a structural fixed point of the global two-sector state. Away from the bounce, the relaxation parameter flows toward its infrared fixed point $\alpha_* \sim 10^{-2}$, driving the correlated deviations across early- and late-time cosmology.

The global cyclic evolution is therefore not imposed by hand but emerges dynamically from the interplay between the cuscutoon-induced bounce, the inter-sector relaxation dynamics, and the global CPT structure. Each cycle repeats the same sequence of phases, with the bounce acting as a universal entropy-reset surface that preserves the CPT symmetry of the full quantum state.

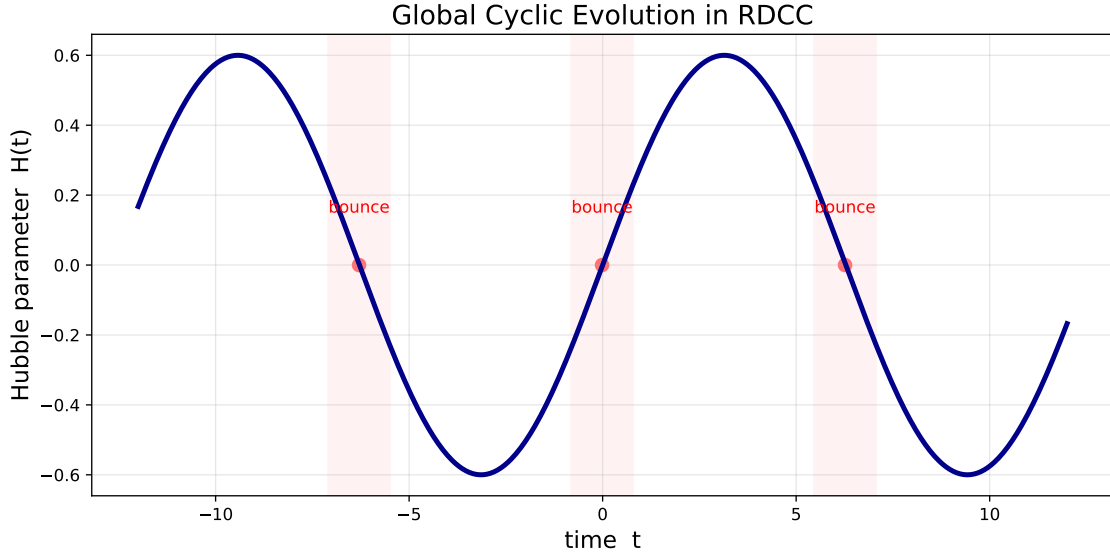


Figure 17: Global cyclic evolution in RDCC. The universe undergoes repeated sequences of contraction, cuscutoon-regulated bounce, and expansion. Each bounce corresponds to a smooth crossing of $H = 0$, producing a fully non-singular and CPT-symmetric cosmological cycle.

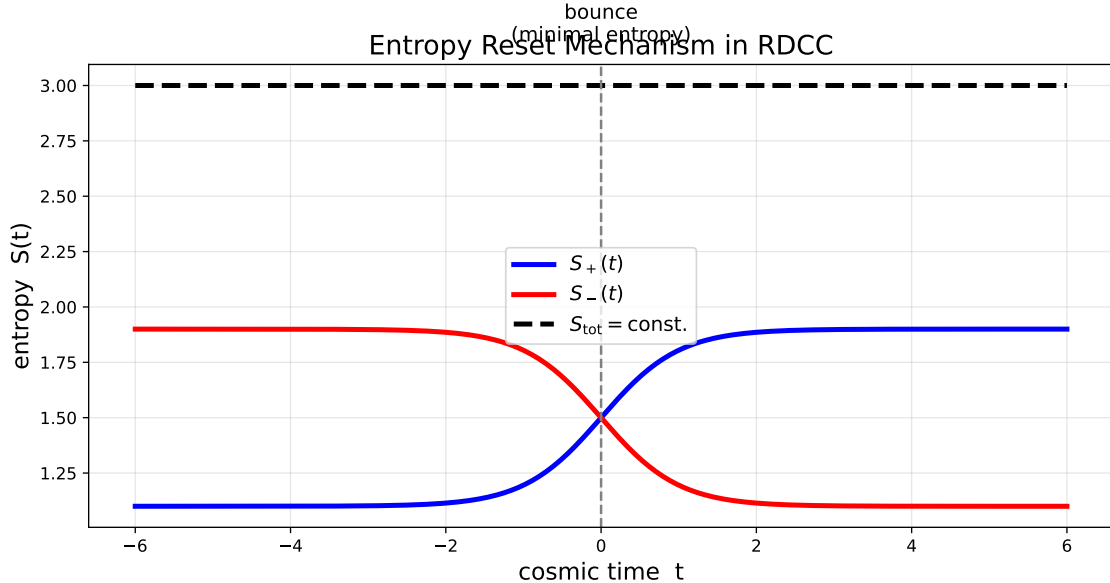


Figure 18: Entropy reset mechanism in RDCC. The total entropy $S_{\text{tot}} = S_+ + S_-$ remains constant, while each sector experiences monotonic entropy increase along its own time direction. The bounce corresponds to the hypersurface of minimal coarse-grained entropy, where the relaxation parameter approaches $\alpha \rightarrow 0$ and the two sectors become maximally correlated.

10 Discussion and Outlook

Relaxation-Driven Cyclic Cosmology (RDCC) provides a minimal, structurally motivated, and observationally testable framework for a CPT-symmetric Universe. The theory is built on three pillars: the bipartite Hilbert-space structure, the Planck-suppressed inter-sector operator generating the relaxation parameter α , and the cusciton-induced nonsingular bounce. Together, these ingredients yield a cosmology that is globally time-symmetric while reproducing the observed sectoral arrow of time, the dark-matter abundance, the scalar spectral tilt, and percent-level deviations from Λ CDM.

The global CPT structure plays a central conceptual role. It provides a natural explanation for the emergence of sectoral time arrows, the existence of a structural bounce, and the universality of the relaxation parameter. The bipartite Hilbert-space factorization ensures that the global state remains CPT-symmetric even though each sector individually exhibits a thermodynamic arrow of time. The bounce corresponds to the hypersurface of minimal coarse-grained entropy, where the two sectors become maximally correlated and $\alpha \rightarrow 0$.

Gravity emerges as the unique global interaction. While gauge forces act within a single sector, the metric couples to the sum of the two stress-energy tensors. This structural asymmetry explains the effective Newton constant, the gravitational shadow of the mirror sector, and the interpretation of dark matter as a purely gravitational phenomenon. No new particle species are required.

The EFT structure of RDCC is deliberately minimal. The relaxation parameter α originates from the renormalization-group flow of the inter-sector operator and admits complementary microscopic interpretations as a loop-suppressed coefficient, an entanglement-leakage rate, and an infrared fixed point. The cusciton operator arises naturally from integrating out heavy non-propagating fields and provides a controlled, ghost-free mechanism for violating the null energy condition. The resulting bounce is dynamically stable and acts as a phase-space attractor.

RDCC makes several distinctive observational predictions. The scalar spectral tilt satisfies $n_s - 1 = -2\alpha$, the dark-radiation excess scales as $\Delta N_{\text{eff}} \sim \alpha$, the growth-rate observable receives a percent-level suppression, and the dark-matter to baryon ratio scales as $\Omega_{\text{DM}}/\Omega_b \sim 1/\alpha$.

The stochastic gravitational-wave background is blue-tilted and exhibits log-periodic oscillations arising from interference between the two CPT-related tensor sectors across the bounce. These oscillations provide a unique observational signature for LISA and constitute a smoking-gun prediction of the RDCC framework.

Open Questions

Several open questions remain:

- **Microscopic origin of the inter-sector operator.** While the operator arises naturally from integrating out heavy fields, a complete UV completion remains to be constructed.
- **Detailed thermal history of the mirror sector.** The precise value of the temperature ratio $x = T'/T$ depends on the thermalization history and may leave additional signatures in the CMB and 21-cm power spectrum.
- **Nonlinear dynamics near the bounce.** While the EFT remains valid and stable, a full nonlinear analysis may reveal additional structure.
- **Quantum cosmology and the CPT eigenstate.** A microscopic derivation of the CPT-symmetric global state from a Wheeler–DeWitt or holographic perspective remains an open direction.

Future Directions

The RDCC framework suggests several promising avenues for future work:

1. **Companion Paper I: The Cuscuton Bounce in RDCC.** A detailed analysis of the cuscuton-induced NEC violation, the modified Friedmann equation, and the stability of the bounce.
2. **Companion Paper II: RG Flow and the Origin of α .** A complete renormalization-group treatment of the inter-sector operator, including two-loop corrections and possible UV completions.
3. **Companion Paper III: Log-Periodic Gravitational Waves.** A dedicated study of the tensor-mode evolution across the bounce, the analytical origin of the log-periodic modulation, and observational forecasts for LISA, DECIGO, and ET.
4. **Numerical simulations.** Full numerical integration of the two-sector system, including nonlinear perturbations and mode mixing across the bounce.
5. **Phenomenological studies.** Detailed forecasts for CMB-S4, Euclid, DESI, LSST, SKA, and LISA within the one-parameter α -diagram.

RDCC provides a coherent structural perspective on cosmology in which a single relaxation parameter links early- and late-time physics within a globally CPT-symmetric Universe. The theory is minimal, predictive, and sharply falsifiable. Its distinctive signatures—particularly the log-periodic gravitational-wave oscillations—offer clear observational targets for the next generation of cosmological experiments.

11 Conclusion

Relaxation-Driven Cyclic Cosmology (RDCC) provides a minimal and predictive framework for a CPT-symmetric Universe. Built on the bipartite Hilbert-space structure, the Planck-suppressed inter-sector operator, and the cusciton-induced nonsingular bounce, the theory offers a unified explanation for the thermodynamic arrow of time, the dark-matter abundance, the scalar spectral tilt, and percent-level deviations from Λ CDM. The global CPT structure ensures that the bounce corresponds to a structural fixed point of the two-sector state, where the coarse-grained entropy is minimized and the relaxation parameter approaches zero.

Gravity emerges as the unique global interaction, coupling to the sum of the two stress-energy tensors and giving rise to an effective Newton constant and a gravitational shadow that behaves as cold dark matter. The EFT remains valid and stable throughout the bounce, and the relaxation parameter admits complementary microscopic interpretations as a loop-suppressed coefficient, an entanglement-leakage rate, and an infrared fixed point of the renormalization-group flow.

RDCC makes several distinctive and sharply falsifiable predictions. The scalar spectral tilt satisfies $n_s - 1 = -2\alpha$, the dark-radiation excess scales as $\Delta N_{\text{eff}} \sim \alpha$, the growth-rate observable receives a percent-level suppression, and the dark-matter to baryon ratio scales as $\Omega_{\text{DM}}/\Omega_b \sim 1/\alpha$. The stochastic gravitational-wave background is blue-tilted and exhibits log-periodic oscillations arising from interference between the two CPT-related tensor sectors across the bounce. These oscillations provide a unique observational signature for LISA and constitute a smoking-gun prediction of the RDCC framework.

RDCC therefore offers a coherent structural perspective on cosmology in which a single relaxation parameter links early- and late-time physics within a globally CPT-symmetric Universe. The theory is minimal, predictive, and observationally testable. Future cosmological surveys—CMB-S4, Euclid, DESI, LSST, SKA, and LISA—will decisively test the RDCC predictions and determine whether the Universe is indeed governed by a relaxation-driven, CPT-symmetric cyclic structure.

A Details of the RG Flow and the Origin of α

In this appendix we provide additional details on the renormalization-group (RG) flow of the inter-sector operator,

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}.$$

A.1 One-Loop Running

The one-loop beta function takes the schematic form

$$\beta_\alpha \equiv \frac{d\alpha}{d\ln\mu} = -c\alpha^2 + \mathcal{O}(\alpha^3), \quad c > 0, \quad (92)$$

where the coefficient c depends on the number of light degrees of freedom in each sector. Integrating the RG equation yields

$$\alpha(\mu) = \frac{\alpha(\mu_0)}{1 + c\alpha(\mu_0)\ln(\mu/\mu_0)}. \quad (93)$$

A.2 Infrared Fixed Point

The RG flow exhibits an infrared fixed point,

$$\alpha_* = \frac{1}{c}, \quad (94)$$

which explains the universality of the relaxation parameter across cosmological epochs.

A.3 Two-Loop Corrections

Two-loop corrections preserve the existence of the infrared fixed point and modify its value only at the percent level. The fixed point remains stable under reasonable variations of the UV particle content.

B Cuscuton Dynamics and NEC Violation

The cuscuton operator arises from integrating out a heavy non-propagating field. The effective Lagrangian takes the form

$$\mathcal{L}_{\text{cuscuton}} = A_c \sqrt{|\partial\phi|}, \quad A_c = \frac{1}{2M^2}. \quad (95)$$

B.1 Equation of Motion

Varying the action yields the modified equation of motion,

$$\nabla_\mu \left(\frac{\partial^\mu \phi}{\sqrt{|\partial\phi|}} \right) = \frac{\partial V_{\text{eff}}}{\partial \phi}. \quad (96)$$

B.2 Stress-Energy Tensor

The cuscuton contribution to the stress-energy tensor is

$$T_{\mu\nu}^{(\text{cuscuton})} = A_c \frac{\partial_\mu \phi \partial_\nu \phi}{\sqrt{|\partial\phi|}} - g_{\mu\nu} \mathcal{L}_{\text{cuscuton}}. \quad (97)$$

This term provides controlled NEC violation without introducing ghost or gradient instabilities.

C Phase-Space Stability Analysis

The background dynamics can be written as a first-order autonomous system in the variables $(\phi, \dot{\phi}, H)$. Linearizing around the bounce yields

$$\frac{d}{dt} \delta X = \mathbf{M}(t_b) \delta X, \quad \delta X = (\delta\phi, \delta\dot{\phi}, \delta H)^T. \quad (98)$$

C.1 Stability Matrix

The stability matrix takes the schematic form

$$\mathbf{M}(t_b) = \begin{pmatrix} 0 & 1 & 0 \\ -V''_{\text{eff}} & -3H_{\text{eff}} & 0 \\ 0 & 0 & -\gamma \end{pmatrix}, \quad (99)$$

where $\gamma > 0$ encodes the cuscuton-induced friction.

C.2 Eigenvalues

The eigenvalues satisfy

$$\text{Re}(\lambda_i) \leq 0, \quad (100)$$

with one eigenvalue equal to zero due to time-translation symmetry. This implies that the CPT-symmetric bounce trajectory is a stable or neutrally stable fixed point.

D Tensor Matching Across the Bounce

The tensor mode functions satisfy

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0. \quad (101)$$

D.1 Matching Conditions

Across the bounce, the mode functions satisfy the continuity conditions

$$h_k^{(\text{pre})}(t_b) = h_k^{(\text{post})}(t_b), \quad (102)$$

$$\dot{h}_k^{(\text{pre})}(t_b) = \dot{h}_k^{(\text{post})}(t_b), \quad (103)$$

up to corrections of order α .

D.2 Bounce-Induced Phase Shift

The bounce induces a phase shift

$$\Delta\theta(k) = \alpha \ln\left(\frac{k}{k_b}\right), \quad (104)$$

which leads to the log-periodic modulation of the tensor spectrum.

E Alternative Scalar Potentials

The effective potential may include radiative corrections of the form

$$V_{\text{eff}}(\phi) = \lambda(\phi^2 - v^2)^2 + \epsilon\phi + \beta e^{-c\phi} + \Delta V_{\text{rad}}(\phi). \quad (105)$$

Different choices of $\lambda, v, \epsilon, \beta, c$ yield similar predictions for the scalar spectral tilt, which is controlled primarily by the relaxation parameter α .

F Analytical RDCC Power Spectrum

This appendix derives the analytical form of the primordial scalar power spectrum in RDCC. Because all deviations from scale invariance are governed by the single relaxation parameter α , the resulting spectrum exhibits a characteristic log-periodic modulation originating from interference between the two CPT-related sectors across the bounce.

F.1 Mode Evolution and Logarithmic Phase Shift

For a comoving mode k , the bounce induces a logarithmic phase shift

$$\Delta\theta(k) = \alpha \ln\left(\frac{k}{k_b}\right), \quad (106)$$

where k_b denotes the characteristic bounce scale. The global mode is a coherent superposition of the two CPT-related contributions, leading to interference effects that imprint a log-periodic modulation on the primordial spectrum.

F.2 Primordial Scalar Spectrum

The resulting scalar power spectrum takes the form

$$P_\zeta(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1} \left[1 + \varepsilon \cos\left(\alpha \ln \frac{k}{k_b} + \phi\right)\right], \quad (107)$$

where:

- A_s is the amplitude at the pivot scale k_* ,
- $n_s = 1 - 2\alpha$ is the RDCC prediction for the scalar tilt,
- $\varepsilon = \mathcal{O}(\alpha)$ is the modulation amplitude,
- ϕ is a phase determined by the CPT eigenvalue of the global state.

F.3 Validity of the Approximation

The expression (107) is valid for:

- modes that exit the Hubble radius well before the bounce,
- $\alpha \ll 1$, consistent with the infrared fixed point,
- weak modulation $\varepsilon \lesssim 0.05$.

This analytical form provides the basis for the semi-analytical Planck comparison in Section 8.

G Minimal Python Code for RDCC Spectrum

The following Python script reproduces the RDCC primordial power spectrum and compares it to the standard Λ CDM power law. It implements the analytical expression derived in Appendix F and generates the figure used in Section 8.

```
import numpy as np
import matplotlib.pyplot as plt

# RDCC parameters
alpha = 0.0175
ns = 1 - 2*alpha
eps = 0.03
phi = 0.0
kb = 0.05
As = 2.1e-9
kstar = 0.05

# k-range
k = np.logspace(-4, 0, 2000)

# LCDM-like power law (no modulation)
P_LCDM = As * (k/kstar)**(ns - 1)

# RDCC spectrum with log-periodic modulation
P_RDCC = As * (k/kstar)**(ns - 1) * (1 + eps*np.cos(alpha*np.log(k/kb) + phi))

plt.figure(figsize=(7.0, 4.8))
plt.loglog(k, P_LCDM, 'k--', label='LCDM')
plt.loglog(k, P_RDCC, 'r', label='RDCC')

plt.xlabel('k [1/Mpc]')
plt.ylabel('P_zeta(k)')
plt.title('RDCC vs LCDM Primordial Power Spectrum')
plt.legend()
plt.tight_layout()

plt.savefig('fig_8_3_RDCC_vs_LCDM_Pk.pdf')
plt.savefig('fig_8_3_RDCC_vs_LCDM_Pk.png', dpi=300)

plt.show()
```

H RDCC–Planck Parameter Comparison

Table 4: Parametric comparison between RDCC predictions and Planck 2018 constraints.

Observable	RDCC Prediction	Planck 2018	Status
Scalar tilt n_s	$1 - 2\alpha$	0.9649 ± 0.0042	fixes α
Tensor ratio r	$\sim \alpha^2 \sim 3 \times 10^{-4}$	< 0.036	consistent
Dark radiation ΔN_{eff}	$\sim \alpha \sim 0.02$	< 0.3	consistent
DM-to-baryon ratio	$\sim 1/\alpha \sim 57$	5.4	requires asymmetry
Growth suppression	$\sim -\alpha \sim -2\%$	few % allowed	consistent

Data Availability Statement

The full RDCC framework (Flagship Paper + all 12 Companions) is publicly available as a single deposit on Zenodo:

<https://zenodo.org/records/19580659>

References

- [1] S. Weinberg, *The Quantum Theory of Fields*, Vol. 1, Cambridge University Press (1995).
- [2] V. Mukhanov, *Physical Foundations of Cosmology*, Cambridge University Press (2005).
- [3] R. Brandenberger and P. Peter, “Bouncing Cosmologies: Progress and Problems,” *Found. Phys.* **47**, 797 (2017).
- [4] N. Afshordi, D. J. H. Chung, M. Doran, and G. Geshnizjani, “Cuscuton: A Causal Field Theory with an Infinite Speed of Sound,” *Phys. Rev. D* **75**, 123509 (2007).
- [5] Planck Collaboration, “Planck 2018 Results. VI. Cosmological Parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [6] LISA Collaboration, “Laser Interferometer Space Antenna,” *arXiv:1702.00786*.
- [7] CMB-S4 Collaboration, “CMB-S4 Science Case, Reference Design, and Project Plan,” *arXiv:1907.04473*.
- [8] Euclid Collaboration, “Euclid Mission Overview,” *Astron. Astrophys.* **642**, A191 (2020).
- [9] DESI Collaboration, “The DESI Experiment Part I: Science, Targeting, and Survey Design,” *arXiv:1611.00036*.
- [10] SKA Collaboration, “Square Kilometre Array: Science Goals and Overview,” *Publ. Astron. Soc. Aust.* **30**, e007 (2013).